



SPACE - ULTRA300 RH

Preliminary Datasheet DS001001 v1.0.0

Product name: SPACE - ULTRA300 RH

Product version: v1.0.0

Document category: Datasheet

Document ID: DS001001

Document version: v1.0.1

Published date: 2026-07

Document status: Released

Document language: English (original)

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1. General information

1.1 Purpose of this document

NanoXplore, a European fabless company, is proud to offer innovative solutions in FPGA, SoC FPGA and ASICs technologies through a 100% European design and supply chain.

In partnership with ESA, CNES and the European Commission, NanoXplore offers sovereignty and ITAR-free solutions to European institutions and industries for Space, Defense and high-reliability systems.

The ULTRA300 FPGA represents the second generation of NanoXplore's flight-proven Radiation-Hardened by Design (RHBD) architecture. By synergizing advanced hardening-by-design techniques with the 28 nm FD-SOI (Fully Depleted Silicon-On-Insulator) process, the ULTRA300 delivers industry-leading resilience against radiation effects and thermal stress. It is specifically engineered to bridge the gap between high-performance processing and extreme environmental survivability.

ULTRA300 FPGA exists in 3 versions:

- **Space ULTRA300 RH** (Radiation Hardened): fully Space qualified devices available in several qualification grades covering all spatial applications spectrum
- **Commercial ULTRA300**: Industrial grade devices taking advantage of ULTRA300 RHBD architecture for cost optimized Space and Defense applications
- **Bare Die ULTRA300**: package less devices dedicated for SIP integration in custom chips

Space and Commercial versions are available in 27x27mm FBGA484, 45x45mm FBGA1152 and Daisy Chain packages.

The purpose of this document is to describe the technical characteristics and interfaces of the **Space ULTRA300 RH** version.

Commercial and Bare Die ULTRA300 versions are documented in dedicated documents. Please refer to the ULTRA300 Documents chapter for the corresponding reference.

1.2 ULTRA300 RH Grades

Space ULTRA300 RH is available in different grades covering all space application spectrum. [ULTRA300 RH Grades \[tab 1 - chap 1\]](#) lists available grades.

Table 1: ULTRA300 RH Grades [tab 1 - chap 1]

Grade	Description
Prototype	No qualification process
Engineering model	Limited qualification process
Equivalent Flight Model ESCC9030	ESCC9030 equivalent qualified devices: simplified qualification process validated by ESA and CNES
Flight Model ESCC9030	Fully ESCC9030 qualified devices
Daisy Chain	Dummy device (not functional), intended to be used for PCB soldering process validation (continuity tests)

2. ULTRA300 Documents

ULTRA300 architecture and features are detailed in different documents listed below.

ULTRA300 Datasheets:

- Space ULTRA300 RH Datasheet

ULTRA300 Datasheets files:

- ULTRA300 Power Estimator
- ULTRA300 Pinout Table
- ULTRA300 CAD Libraries archive
- ULTRA300 Daisy Chain archive

ULTRA300 User Guides:

- ULTRA300 Fabric User Guide
- ULTRA300 Configuration Guide
- ULTRA300 Eval Kit Guide

3. Summary

This chapter provides a summary of ULTRA300 key features.

Radiation Tolerance

- Radiation-hardened design applied to configuration memories and registers for enhanced resilience
- Single Event Upset (SEU) immunity up to LET > 60 MeV.cm²/mg
- Total Ionizing Dose (TID) tolerance up to 50 Krads (Si), ensuring long-term reliability in harsh environments
- Embedded Error Detection and Correction (EDAC) mechanisms safeguard user memory integrity
- TMR (Triple Modular Redundancy) SPI configuration mode

Main Features

- Technology Node: STM 28 nm FD-SOI process technology
- 12-bit ADC @ 5 Mbps, 10-bit DAC @ 2Mbps
- Programmable Logic: ~ 300K 4-Input Look-up tables
- Arithmetic Capabilities: DSP Blocks for complex arithmetic operations
- User-configurable memories with variable width and depth
- Configuration modes: JTAG, 16 bits Parallel, Master SPI/QSPI (optional TMR), Slave SPI, SpaceWire
- Integrated SpaceWire interfaces available for user applications
- Dedicated low-skew distribution network for clock, reset and load enables signals
- On-chip thermal monitoring for system reliability
- Security feature: AES-256 Bitstream encryption. Readback, anti-tampering and anti-rollback protections

Input / Output Features

- Multi voltage IO support from 1.2V to 3.3V
- Cold sparing support
- Programmable output drive to support multiple industry standards
- Embedded logic to support DDR2
- Up to 1.6 Gbps maximum IO support for SSTL and HSTL standards
- LVDS compatible mode up to 800 MHz
- All pins support 2000V of ESD-HBM
- Programmable delay lines on complex IO pins
- Programmable resistive termination on complex IO pins
- Optional single rate register on direct IO pins and single or double rate register on complex IO pins

4. Device Overview

ULTRA300 FPGA is a Radiation Hardened By Design (RHBD) SRAM-based FPGA manufactured on STM 28 nm FD-SOI Space process.

This chapter provides an overview of ULTRA300 architecture and its key features.

4.1 Resources

Table 2: Device Resources [tab 1 - chap 4]		
Device	ULTRA300 Package 484 pins	ULTRA300 Package 1152 pins
Logic Modules		
Register		273 408
LUT-4		290 496
Carry		68 352
Embedded RAM		
Core RAM Blocks (48K-bit)		448
Core RAM Bits (K = 1024)		21 504 K
Core Register File Blocks (32 x 18-bit)		1 424
Core Register File Bits (K = 1024)		801 K
Total RAM Bits (K = 1024)		22 305 K
Additional Features		
Embedded DSP (24 x 18-bit)		896
ADC (12-bit @ 5Mbps)		4 ⁽¹⁾
DAC (10-bit @ 2Mbps)		4 ⁽¹⁾
SPI Controller		3
IO Resources		
User IO	239	500
IO Complex Banks (34 IOs per bank)	4	10

Device	ULTRA300	ULTRA300
	Package 484 pins	Package 1152 pins
IO Direct Banks	1 small	1 small
(small = 16 IOs per bank, large = 24 IOs per bank)	4 large ⁽²⁾	6 large
Service IO	48 ⁽³⁾	57
Embedded Serial Link		
400Mbps TX/RX SpaceWire	8	20
(2 per Complex Bank)		
HSSL, Multi-protocol SERDES TX/RX	4	16
Packages		
Available Packages	FBGA484	FBGA1152

- (1)** Direct Banks and ADC/DAC share IOs in FBGA484 package.
- (2)** IOB08 has only the first 15 pads connected in FBGA484 package.
- (3)** Some Service Bank IOs are not connected in FBGA484 package.

4.2 Device Architecture

ULTRA-300 FPGA is a 28 nm programmable chip composed of a central fabric embedding the programmable logic, RAM and DSP blocks, and IO peripherals as user IOs, HSSL, ADCs and DACs.

The programmable logic resources are arranged in hierarchical structures called TILE and CGB (Coarse Grain Block) with a specific local interconnect network.

The IO peripherals are arranged in multiple banks and are located in the die ring.

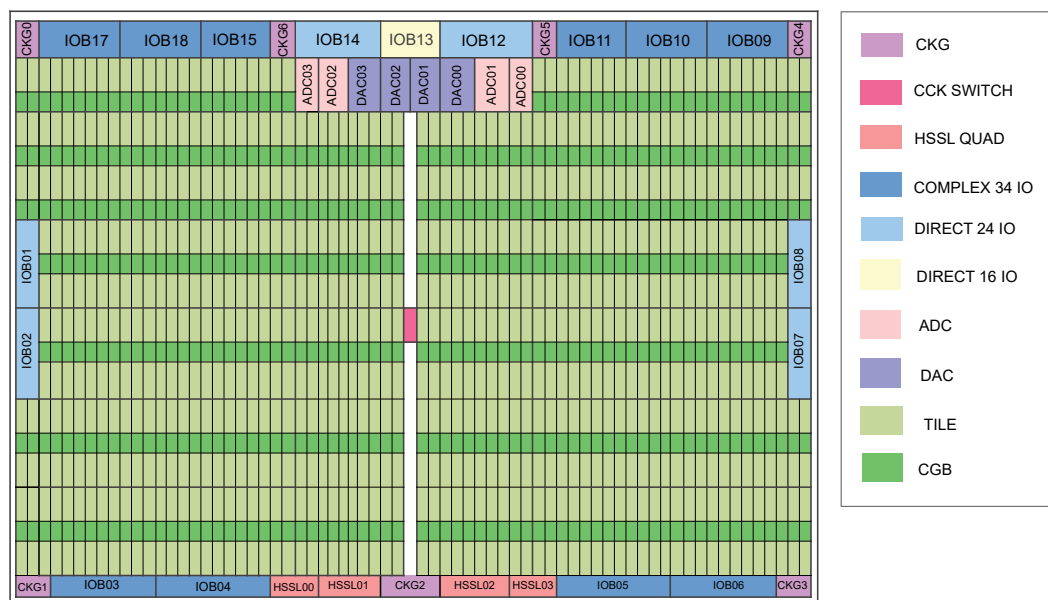


Figure 1: ULTRA300 die device floorplan [fig 1 - chap 4]

Note:

- Some ring die instances are not connected to the *FBGA484* package.
- HSSL is currently under characterization – HSSL feature will be described in a later update of the Datasheet

4.3 Die Features

The ULTRA300 FPGA fabric includes a matrix of TILE and CGB rows, interconnect resources and an IO ring.

ULTRA300 hardware capacities are composed of:

- **An array of 11 rows of TILE (712 TILE in total):**
- 273 408 4-input LUTs
- 68 352 Carry Logic bits
- 1424 Register files

- **An array of 7 rows of CGB (448 CGB in total):**
 - 1 x 48 Kbit True Dual Port RAM blocks per CGB (total 448 DPRAM blocks)
 - 2 x DSP blocks per CGB (total of 896 DSP)
- **An IO Ring with the following characteristics:** (FBGA484 package has less resources)
 - Direct IO banks:
 - 6 x 24 IO bank: 2 top, 2 Left and 2 Right
 - 1 x 16 IO bank: 1 top
 - 10 Complex IO Banks (34 IOs per bank): 6 top, 4 bottom
 - ADC / DAC
 - 4 x ADC: 12-bit @ 5Mbps
 - 4 x DAC: 10-bit @ 2Mbps
 - 1 Service bank of 57 IO Bank Prog/interface
 - 4 HSSL (High Speed Serial Link) blocks with block including 4 RX/TX Transceivers
- **7 Clock generators (CKG):** CKG0, CKG1, CKG2, CKG3, CKG4, CKG5, and CKG6
 - Each CKG includes 1 PLL and 10 WFG

4.4 Radiation Tolerance

All resources are protected against radiation through the following measures:

- FD-SOI technology, which offers excellent intrinsic radiation tolerance
- Configuration memory cells, user flip-flops and user register file blocks are designed using dedicated radiation hardening techniques to ensure immunity against Single Event Effect (SEE) to cover all possible effects (upset, transient, latch-up)
- Embedded dual-port RAM is safeguarded with Error Correction Code (ECC)
- Critical logic blocks are triplicated to enhance fault tolerance and reliability

4.5 Security Features

The ULTRA300 integrates OTP (One Time Programmable) memories to store encryption keys and allows AES-256 bitstream encryption.

The ULTRA300 integrates also anti-tampering and anti-rollback protection mechanisms.

4.6 Configuration

4.6.1 Configuration Modes

The ULTRA300 configuration is controlled by the BSM (Bitstream Manager) component. ULTRA300 is configured by loading the bitstream into internal configuration memory from an external memory or device and using dedicated IO interfaces:

- JTAG: always available regardless of the selected mode
- 3 x SPI: supports sequential and TMR. Supports SPI modes SPI, DUAL-SPI and QSPI
- SpaceWire(1)
- 16b parallel

(1) Dedicated SpaceWire configuration interface is not connected for 484-pins

4.6.2 Bitstream Size

The ULTRA300 NX2H300TSA bitstream size depends on the application size (configuration) and the number of user Core RAM and Core Register File to be initialized.

Table 3: Bitstream size [tab 2 - chap 4]	
Element	Size (Mb)
Maximum Configuration (100%)	52.353
Maximum D-Flip-Flop initialization (273408)	0.279 (1.02b/DFF)
Maximum Core RAM initialization (448)	44.040 (98.30Kb/RAM)
Maximum Core Register File initialization (1424)	2.187 (3.07Kb/2RFB)
Maximum Configuration Memory Integrity Check	0.089
Maximum total	98.948

Most applications do not require to initialize all instances and bitstream will be reduced.

5. Electrical Specifications

5.1 Operating Conditions

Table 4: Device Operating Condition [tab - chap 5]

Grade	Parameter	Value
Prototype	Temperature Range	+25°C
Engineering model	Temperature Range	-40°+125°C
Equivalent Flight Model ESCC9030	Temperature Range	-40°+125°C
Flight Model ESCC9030	Temperature Range	-40°+125°C
Daisy Chain	Temperature Range	+25°C

5.2 Power supplies

ULTRA300 FPGA requires specific power supplies, as detailed in the following sections.

5.2.1 Absolute Maximum Ratings

Table 5: Absolute Maximum Ratings [tab 2 - chap 5]

Type	Symbol	Description	Minimum	Maximum	Units
Common	VDDCORE	Core logic supply	-0.5	1.1	V
	VDD1V8A	Auxiliary analog supply	-0.5	1.98	V
Direct IO bank	VDDIO_x	IO banks supplies	-0.5	3.63	V
Complex IO bank	VDDIO_x	IO banks supplies	-0.5	1.98	V
	VTO_x	Termination supplies	IO Standard dependent	IO Standard dependent	V
	VDDS_x	On-Chip Termination buffer switch supply	-0.5	1.98	V
Service IO Bank	VDDIO_SERVICE	Service IO Bank supply	-0.5	3.63	V
Bank current estimation	lin	Maximum current through an IO when forward biasing the clamp diode (the bank is powered or unpowered)	-	10	mA
HSSL ⁽¹⁾	HSSLx_TXx_VDDA	HSSL TX driver analog supply	TBD ⁽¹⁾	TBD ⁽¹⁾	V
	HSSLx_VDD1V8A	HSSL RX/TX analog supply	TBD ⁽¹⁾	TBD ⁽¹⁾	V
	HSSLx_VDDCORE	HSSL digital supply	TBD ⁽¹⁾	TBD ⁽¹⁾	V
CKGEN	CKGENx_PLL_VDD1V8A	PLL auxiliary analog supply	-0.5	1.98	V
	CKGENx_PLL_VDDA	PLL analog supply	-0.5	1.1	V
ADC	ADC_VDD1V8A	ADC analog supply	-0.5	1.98	V
DAC	DAC_VDD1V8A	DAC analog supply	-0.5	1.98	V

⁽¹⁾ HSSL is currently under characterization – final values will be shared in a later update of the Datasheet

5.2.2 Recommended Operating conditions

Table 6: Recommended Operating conditions [tab 3 - chap 5]

Power Supply Type	Power supply	Description	Value (V)
Common	VDDCORE	Core logic supply	1.0 ± 5%
	VDD1V8A	Auxiliary analog supply	1.8 ± 5%
Direct IO bank	VDDIO_x ⁽¹⁾	IO banks supplies	1.8, 2.5 or 3.3 ± 10%
Complex IO bank	VDDIO_x	IO banks supplies	1.2, 1.5 or 1.8 ± 5%
	VTO_x	Termination supplies	IO Standard dependent, (Please refer to IO Standard DC characteristics [tab 6 - chap 5])
	VDDS_x	On-Chip Termination buffer switch supply	1.8 ± 5%
Service IO Bank	VDDIO_SERVICE	Service IO Bank supply	3.3 ± 10%
HSSL ⁽²⁾	HSSLx_TXx_VDDA	HSSL TX driver analog supply	TBD ⁽²⁾
	HSSLx_VDD1V8A	HSSL RX/TX analog supply	TBD ⁽²⁾
	HSSLx_VDDCORE	HSSL digital supply	TBD ⁽²⁾
CKGEN	CKGENx_PLL_VDD1V8A	PLL auxiliary analog supply	1.8 ± 5%
	CKGENx_PLL_VDDA	PLL analog supply	1.0 ± 5%
ADC	ADC_VDD1V8A	ADC analog supply	1.8 ± 5%
DAC	DAC_VDD1V8A	DAC analog supply	1.8 ± 5%

⁽¹⁾ In FBGA484, there is no bank voltage limitation when ADC/DAC are used. Shared Direct IO banks can operate at a different voltage (1.8, 2.5 or 3.3V) than ADC/DAC blocs (1.8V). Power sequences shall comply with the recommendations provided in section “Power and Reset Sequencing”, to ensure correct device operation.

⁽²⁾ HSSL is currently under characterization – final values will be shared in a later update of the Datasheet

5.2.3 Unused Power Supply Recommendations

Unused power supplies may be left floating or tied to ground.

This recommendation applies to:

- Complex and Direct IO bank voltages
- ADC and DAC IO bank voltages
 - See additional recommendations below FBGA484 package
- HSSL IO bank voltages

FBGA484 package only:

When a Direct IO bank shared with ADC/DAC is used (IOB12, IOB13 and/or IOB14), ADC/DAC power supplies in the shared bank must be powered even if the ADC/DAC is not used.

Ex:

- IO used in IOB12: IOB12_VDDIO and ADC_VDD1V8A must be powered
- IO used in IOB14: IOB14_VDDIO and ADC_VDD1V8A must be powered
- IO used in IOB13: IOB13_VDDIO and DAC_VDD1V8A must be powered

5.2.4 Power Monitoring Pin

ULTRA300 powering status can be monitored using “Power OK” pin located in IO Bank Service (“POK” in pinout table).

Power OK value is driven by a logic AND on VDD1V8A and VDDCORE power supply. Power OK output is driven by VDDIO_SERVICE power supply.

Power OK output generates a valid high level when both supplies (VDD1V8A and VDDCORE) are above the threshold and VDDIO_SERVICE is in valid operating condition.

Table 7: Power OK management [tab 4 - chap 5]

Pin	Power supply	Thresholds
Power OK	VDD1V8A	On ~1.4V, Off ~0.95V
	VDDCORE	On ~0.8V, ~0.55V
	VDDIO_SERVICE	-

Power OK pin is not available in the 484 pins package. An external Power OK status can be generated by doing a logic AND on VDD1V8A, VDDCORE and VDDIO_SERVICE.

5.3 IO Characteristics and supported IO Standards

Each pad of each IO bank can be configured as input-only, output-only or input-output.

5.3.1 Acronyms

Table 8: Electrical Acronyms [tab 5 - chap 5]

Acronym	Description
VIH/VIL	Minimum/Maximum required voltage level at the input to detect high/low logic level
VOH/VOL	Minimum/Maximum voltage level at the output to drive high/low logic level
VINP	Input voltage level
VICM	Input common mode voltage level in differential signaling
VID	Differential voltage value at the input
VOCM	Output common mode voltage level in differential signaling
VOD	Differential voltage value at the output
VTO	Voltage Termination Output

5.3.2 IO Standards DC AC Specifications (Direct and Complex IO)

Table 9: IO Standard DC characteristics [tab 6 - chap 5]

VDDIO (V)	Vref (V) ⁽¹⁾ (Internally generated)						VTO (V)			Standard Support
	MIN	TYP	MAX	MIN	TYP	MAX	MIN	TYP	MAX	
LVC MOS3V3	3.0	3.3	3.6	-	-	-	-	-	-	JESD8C.01
LVC MOS2V5	2.3	2.5	2.7	-	-	-	-	-	-	JESD8-5A
LVC MOS1V8	1.65	1.8	1.95	-	0.9	-	-	-	-	JESD8-7A
LVC MOS1V5	1.4	1.5	1.6	-	0.75	-	-	-	-	JESD8-11A
LVC MOS1V2	1.1	1.2	1.3	-	0.6	-	-	-	-	JESD8-12A
LVDS1V8 ⁽²⁾	1.71	1.8	1.89	-	-	-	-	-	-	
SSTL 1V8 class I/II	1.7	1.8	1.9	0.825	0.9	0.975	(0.5*VDDIO)-0.04	0.5*VDDIO	(0.5*VDDIO)+0.04	JESD8-15
SSTL 1V5 class I/II	1.425	1.5	1.575	0.7	0.75	0.8	0.49*VDDIO	0.5*VDDIO	0.51*VDDIO	JESD79-3F
SSTL 1V2 class I/II	1.14	1.2	1.26	0.50	0.6	0.65	0.49*VDDIO	0.5*VDDIO	0.51*VDDIO	-
HSTL 1V8 class I/II	1.7	1.8	1.9	0.825	0.9	0.975	-	Vref	-	JESD8-6
HSTL 1V5 class I/II	1.4	1.5	1.6	0.675	0.75	0.9	-	Vref	-	JESD8-6

⁽¹⁾ Vref is an internally generated reference voltage used in the input buffer comparator for single-ended IO standards. It is generally equal to VDDIO/2.

⁽²⁾ Pad complex input and output buffers are LVDS 1V8 compatible, therefore have only restricted support for ANSI/TIA/EIA-644 specifications. The voltage value at a given pad should not exceed 1.98V. The user should ensure that none of the emitters connected to a pad will violate this condition.

Table 10: IO Single-Ended Standards AC/DC Input Output Specifications [tab 7 - chap 5]

Standard	AC				DC				IOH (mA) ⁽¹⁾	IOL (mA) ⁽¹⁾	Compatible Standard
	VIH(V)	VIL(V)	VOH(V)	VOL(V)	VIH (V)	VIL (V)	VOH (V)	VOL (V)			
LVC MOS 3V3	-	-	-		2.2	0.7	2.2	0.5	-2, -4,	2, 4, 8,	JESD8C.01
LVC MOS 2V5	-	-	-	-	1.8	0.6	1.6	0.8	-8, -12, -16	12, 16	JESD8-5A
LVC MOS 1V8	-	-	-	-	1.21	0.56	1.16	0.57	-2, -4, -8, -12, -16	2, 4, 8, 12, 16	JESD8-7A
LVC MOS 1V5	-	-	-	-	1.02	0.45	0.97	0.49	-2, -4, -8, -16	2, 4, 8, 16	JESD8-5
LVC MOS 1V2	-	-	-	-	0.84	0.34	0.75	0.41	-2, -4, -6	2, 4, 6	JESD8C-12A
SSTL 1V8 class I	Vref +0.25	Vref-0.25	VTO +0.6	VTO-0.6	Vref+ 0.125	Vref-0.125	Vt+0.6	Vt-0.6	-8.6	8.6	JESD8-15
SSTL 1V8 class II	Vref +0.25	Vref-0.25	VTO +0.6	VTO-0.6	Vref+ 0.125	Vref-0.125	Vt+0.6	Vt-0.6	-13.4	13.4	JESD8-15
SSTL 1V5 class I	Vref +0.15	Vref-0.15	VTO +0.6	VTO-0.6	Vref+ 0.1	Vref-0.1	0.8*VDDIO	0.2*VDDIO	-8	8	JESD79-3E
SSTL 1V5 class II	Vref +0.15	Vref-0.15	VTO +0.6	VTO-0.6	Vref+ 0.1	Vref-0.1	0.8*VDDIO	0.2*VDDIO	-16	16	JESD79-3E
SSTL 1V2	Vref +0.15	Vref-0.15	-	-	Vref+ 0.1	Vref-0.1	0.8*VDDIO	0.2*VDDIO	-	-	-
HSTL 1V8 class I	Vref+0.2	Vref-0.2	VD- DIO-0.6	0.6	Vref+0.1	Vref-0.1	VDDIO-0.5	0.5	-8	8	JESD8-6
HSTL 1V8 class II	Vref+0.2	Vref-0.2	VD- DIO-0.6	0.6	Vref+0.1	Vref-0.1	VDDIO-0.5	0.5	-16	16	JESD8-6
HSTL 1V5 class I	Vref+0.2	Vref-0.2	VD- DIO-0.5	0.5	Vref+0.1	Vref-0.1	VDDIO-0.4	0.4	-8.0	8.0	JESD8-6
HSTL 1V5 class II	Vref+0.2	Vref-0.2	VD- DIO-0.5	0.5	Vref+0.1	Vref-0.1	VDDIO-0.4	0.4	-16	16.0	JESD8-6

⁽¹⁾ 16mA feature is being characterized

Table 11: LVDS Receiver (RX) Characteristics [tab - chap 5]

DC

Standard	Vin (V)	Vin Diff (V) ⁽²⁾	VICM(V)	Input	Input Current	Diff Impedance
	min - max	min - max	min - max ⁽³⁾	Current (A)	Balance (A)	Range (Ω)
LVDS 1V8 ⁽¹⁾	0 – 1.8	0.1 – 1.98	0.3 - 1.3 for Vin Diff ≥ 0.2V 0.5 - 1.3 for Vin Diff ≥ 0.1V	<200u	<2u	100 ± 20%

⁽¹⁾ Pad complex input and output buffers are LVDS 1V8 compatible, therefore have only restricted support for ANSI/TIA/EIA-644 specifications.

⁽²⁾ The voltage value at a given pad should not exceed 1.98V. The user should ensure that none of the emitters connected to a pad will violate this condition.

⁽³⁾ Under recommended operating conditions of supply rails.

⁽⁴⁾ Calculated with 100 Ω differential termination.

Table 12: LVDS Driver (TX) Characteristics [tab - chap 5]

DC

Standard	VOD (V)	VOCM (V)	Ishortcircuit (A)	Ileakage (A)	Trise/Tfall (ps)
	min - max ⁽²⁾	min - max ⁽²⁾			
LVDS 1V8 ⁽¹⁾	0.26 - 0.55	0.58 - 1.05	12m	<1u	140 - 360

⁽¹⁾ Pad complex input and output buffers are LVDS 1V8 compatible, therefore have only restricted support for ANSI/TIA/EIA-644 specifications.

⁽²⁾ Calculated with 100 Ω differential termination.

5.3.3 IO Standards DC AC Specifications (HSSL IO) - preliminary

All HSSL lanes must be AC coupled.

HSSL is currently under characterization. Final values will be shared in a later update of the Datasheet

5.4 Cold Sparing

ULTRA300 FPGA has bidirectional IO pads supporting cold sparing implementation.

The cold sparing feature enables multiple FPGA devices to share the same IO bus. One device operates as the active FPGA, while the others remain unpowered and serve as backup units in case of failure.

When not in use, the backup devices have their VDDIO power pins grounded, ensuring they do not interfere with the bus. If a high logic level is driven on the shared pad, no current flows into the grounded VDDIO of the inactive devices.

During signal transitions, the pad behaves like a small capacitor (a few picofarads), resulting in charging and discharging currents. These transient currents are expected and do not affect the grounded backup devices due to the cold sparing configuration

[Cold-spare complex IO characteristics \[tab 10 - chap 5\]](#) lists cold-spared complex IO Characteristics (all supplies are grounded; pad voltage is equal to VDDIO+10%). Values are shown for the current flowing through Pad pin.

Table 13: Cold-spare complex IO characteristics [tab 10 - chap 5]

Condition	Ipadmaxdc @ TT, -40°C (nA)	Ipadmaxdc @ TT, 25°C (nA)	Ipadmaxdc @ TT, 125°C (nA)
VDDIO 1.2V	0.25	1.75	390
VDDIO 1.5V	1.5	4.5	425
VDDIO 1.8V	1.7	39	590

[Cold-spare direct IO characteristics \[tab 11 - chap 5\]](#) lists cold-spared direct IO Characteristics (all supplies are grounded; pad voltage is equal to VDDIO+10%). Values are shown for the current flowing through Pad pin.

Table 14: Cold-spare direct IO characteristics [tab 11 - chap 5]

Condition	Ipadmaxdc @ TT, -40°C (nA)	Ipadmaxdc @ TT, 25°C (nA)	Ipadmaxdc @ TT, 125°C (nA)
VDDIO 1.8V	0.13	0.3	27
VDDIO 2.5V	0.55	7	112
VDDIO 3.3V	90	350	875

5.5 Reset Management

ULTRA300 reset is controlled by 2 Service IO bank pins:

- RST_SOFT_N
- RST_HARD_N

Both signals must be controlled with respect to sequences and constraints detailed in chapter [Power and Reset Sequencing](#).

5.5.1 RST_HARD_N

RST_HARD_N is the main reset signal. It resets both the Bitstream Manager (BSM) Engine (meaning an FPGA reconfiguration) and the fabric programmable matrix.

All Reset Sequences must respect:

- RST_HARD_N reset length $\geq 3\mu\text{s}$

RST_HARD_N **cannot** be left unconnected.

5.5.2 RST_SOFT_N

RST_SOFT_N is an additional reset signal which resets only the BSM Engine.

It **MUST** be used only in combination with RST_HARD_N. The different use cases of RST_SOFT are:

- RST_SOFT_N = RST_HARD_N Or RST_SOFT_N = Unconnected. As RST_SOFT_N has an internal pull-up, both connections are identical. This mode may propagate glitches to the IO

Or

- In combination with RST_HARD_N to prevent IO glitches:
 - After a RST_HARD_N assertion, at power-up or during a reset sequence, to delay the bitstream configuration and keep the fabric logic in a reset state
 - Before a RST_HARD_N assertion, at power-down or during a reset sequence, to ensure an IO glitch free sequence

5.5.3 ULTRA300 Configuration after a reset operation

ULTRA300 configuration starts when both RST_HARD_N and RST_SOFT_N are high and when the BSM leaves its reset state.

5.6 Power and Reset Sequencing

Several problems may occur when ULTRA300 power and reset sequences are not controlled:

- Transient IO glitches, which may damage external components
- Uncontrolled power consumption peaks, which may not be compatible with power architecture

To ensure stability during power and reset sequences, ULTRA300 must respect specific sequences. The purpose of these sequences is to control the power consumption and set the FPGA in a safe configuration mode to avoid IO glitches.

ULTRA300 power-up phases are shown in [ULTRA300 power-up phases \[fig 1 - chap 5\]](#).

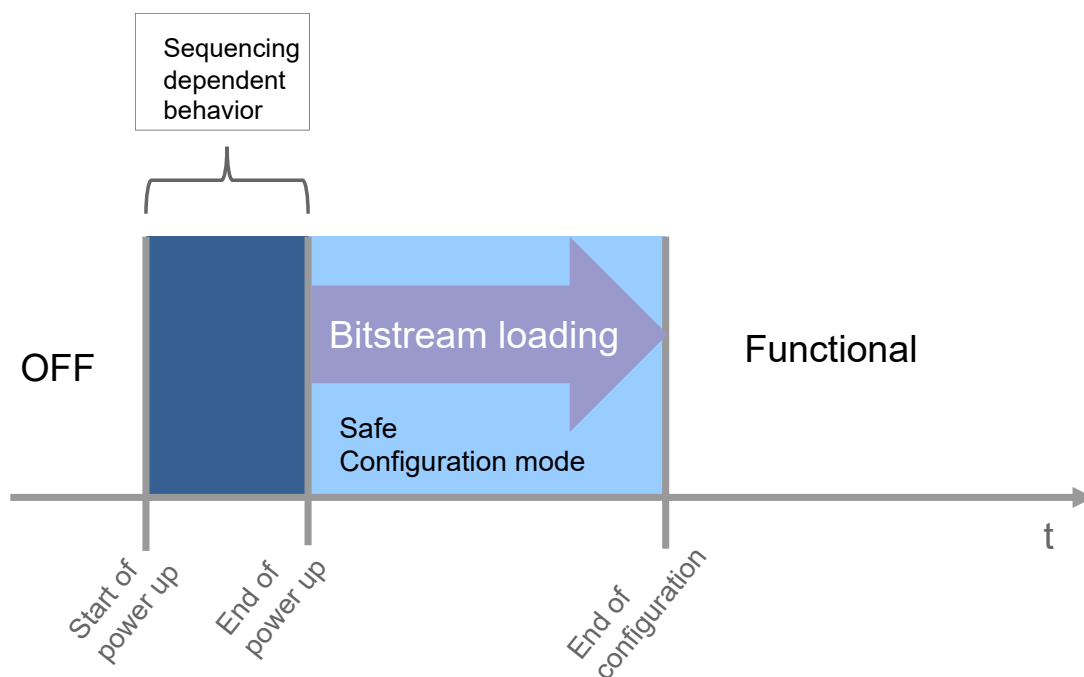


Figure 2: ULTRA300 power-up phases [fig - chap 5]

4 power supplies control the state of ULTRA300 FPGA during powering phases:

- VDD1V8A
- VDDCORE
- VDDIO_SERVICE
- VDDIO_X (represents all user Direct and Complex banks voltages)

As shown in [ULTRA300 Power Sequences \[fig 2 – chap 5\]](#), power sequences and reset sequence must be controlled. The next chapters details specific sequences.

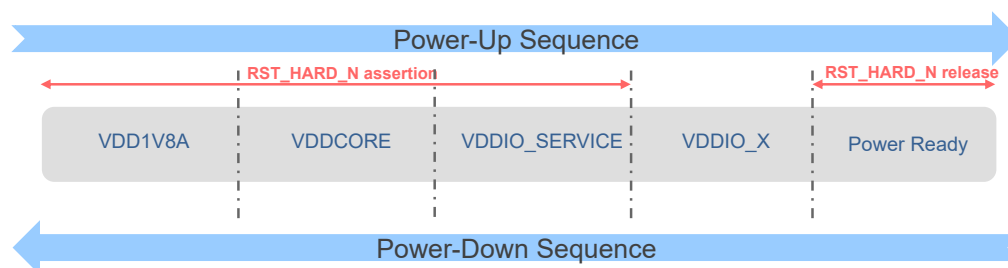


Figure 3: ULTRA300 Power Sequences [fig – chap 5]

5.6.1 Power-up sequence

To ensure safe configuration mode, recommended power sequencing shown in [ULTRA300 power-up sequence \[fig 3 – chap 5\]](#) should be applied. This implies that VDD1V8A, VDDCORE and VDDIO_SERVICE should be electrically isolated and powered up before from other power rails. The RST_HARD_N input must be asserted Low before the power up of user IO Bank voltages (all user IO Bank voltages).

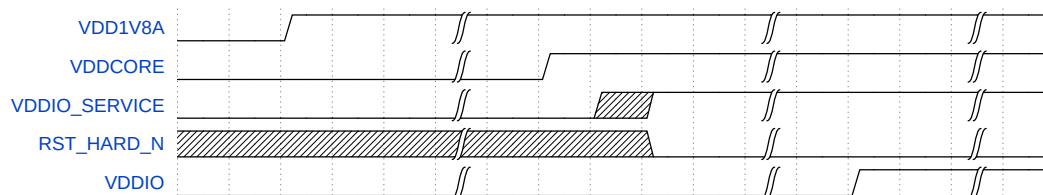


Figure 4: ULTRA300 power-up sequence [fig – chap 5]

There are no defined time delay constraints between the power up of different power rails. A general recommendation for the power-on sequence is to enable each power rail only after the previous rail reaches 90% of its nominal voltage, while avoiding long delays between rails.

All Reset Sequences must respect:

- RST_HARD_N reset length $\geq 3\mu\text{s}$

5.6.2 Power-down sequences

Power-down sequence with Reset

ULTRA300 can be shut down to reach the end of bitstream operability either through a component reset or a direct power-off.

While powering-off with a reset, to ensure a glitch free reset, a soft reset should be applied before initiating a hard reset. The soft reset ensures that the IO returns to Safe-Configuration mode, thereby preventing hard reset from inducing any transient glitch on user IO pins. This recommended reset sequence is shown in [ULTRA300 Recommended Reset Sequence before power-off \[fig 4 - chap 5\]](#). If a shutdown is expected after the reset, the use of recommended reset sequence allows all power rails to be powered off simultaneously without any specific ordering constraints.

CLK_BSM is internally generated and by default at 100MHz unless modified by user during bitstream loading phase. A delay of 100 cycles of CLK_BSM between the RST_SOFT_N and RST_HARD_N is required to ensure entry into Safe Configuration Mode.

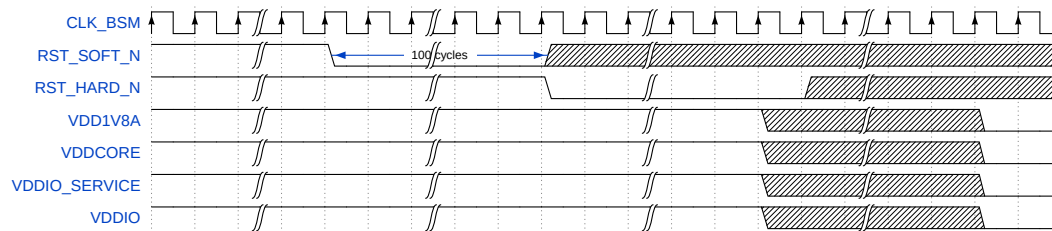


Figure 5: ULTRA300 Recommended Reset Sequence before power-off [fig - chap 5]

All Reset Sequences must respect:

- RST_HARD_N reset length $\geq 3\mu\text{s}$

Power-down sequence without Reset

In cases where a power-off occurs without reset, the recommended power-off sequence shown in [ULTRA300 recommended power-off sequence without reset \[fig 5 - chap 5\]](#) ensures glitch-free shutdown. The power-down sequence is the reverse of the power-on sequence

When respecting this sequence, reset signals do not need to be controlled.

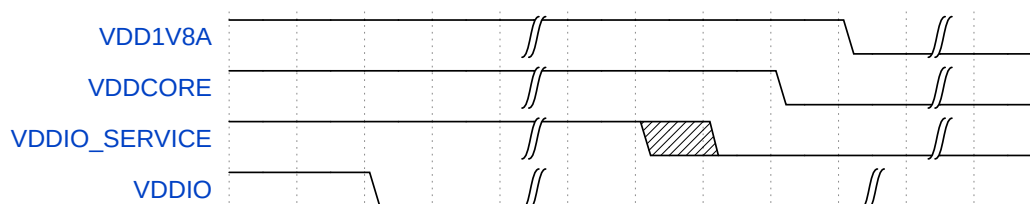


Figure 6: ULTRA300 recommended power-off sequence without reset [fig - chap 5]

5.6.3 Reset Sequence

To prevent spreading some glitches from the IOs to external devices during a hard reset, the READY state must be de-asserted before doing a hard reset to disable IO ring and set all IOs to High-Z with pull-up by using RST_SOFT_N pin.

This reset sequence, detailed in [ULTRA300 Reset Sequence \[fig 6 - chap 5\]](#), must respect timings rules:

- $Tp0 \geq 100 \text{ CLK_BSM periods}$
- $Tp1 \geq 3 \mu\text{s}$
- $Tp2 \geq 20\,000 \text{ CLK_BSM periods}$

CLK_BSM is internally generated and by default at 100MHz unless modified by user during bitstream loading phase.

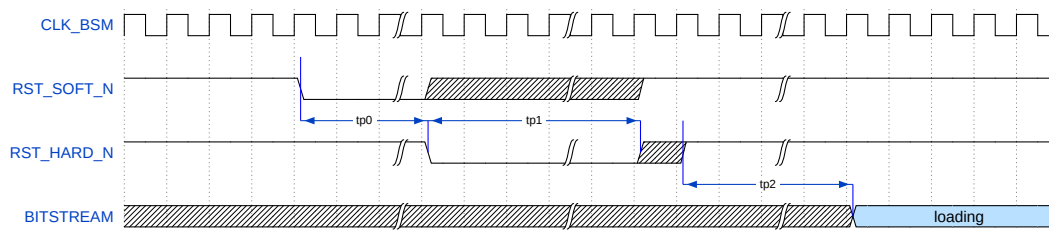


Figure 7: ULTRA300 Reset Sequence [fig - chap 5]

5.7 Service IO Bank IO Protection

Service IO bank regroups both the input pins defining ULTRA300 configuration mode and the configuration interface pins. Depending on the selected mode, some configuration pins can be either an input or an output. All pins in this bank are powered by VDDIO_SERVICE.

Therefore, when VDDIO_SERVICE is being activated, there is a time period where these configurations pins state may differ from intended use. For example, an ULTRA300 Input pin (i.e SPI MISO) may be configured as an output during this short period.

To avoid any risk of damaging external devices, it is strongly recommended to add 22ohm series resistors to any Service IO bank with following characteristics

- Pin connected to an external device (ex: flash memory, any SPI device, etc...)
- External device pin direction = INPUT

5.8 Power Consumption

5.8.1 Power Estimator

To assist PCB designers in properly sizing power supplies, a power estimation tool is available in a separate document.

Please refer to the ULTRA300 Documents chapter for the corresponding reference.

6. Fabric Performance

6.1 PLL

This chapter describes fabric performance.

Table 15: PLL Characteristics [tab 1 - chap 6]

PLL features		
PLL_F _{INMIN} ⁽¹⁾⁽²⁾	Minimum input clock frequency	10 MHz
PLL_F _{INMAX} ⁽¹⁾⁽²⁾	Maximum input clock frequency	50 MHz
PLL_F _{INJITTER}	Maximum input clock period jitter	± 2% @ 10MHz
		± 5% @ 50MHz
PLL_F _{INDUTY}	Input duty cycle range: 10–50 MHz	65% - 35%
PLL_F _{VCOMIN}	Minimum PLL VCO frequency	300 MHz
PLL_F _{VCOMAX}	Maximum PLL VCO frequency	800 MHz
PLL_T _{LOCKMAX}	PLL maximum lock time	100 µs
PLL_F _{OUTMIN}	Minimum PLL frequency	30 MHz
PLL_F _{OUTMAX}	Maximum PLL frequency	800 MHz
PLL_T _{OUTJITTER}	PLL Chip Output Jitter	<2% (Worst-Case clock path)

- (1) PLL input clock frequency range after the input divider
- (2) PLL input clock frequency range can be extended using PLL input divider with factor [1;32]

6.2 Complex IO Banks

6.2.1 IO Complex Standard List

Table 16: IO Complex Standard List [tab 2 - chap 6]

Standard	Type	Supply	Drive ⁽¹⁾	Speed MHz or Mb/s	Special considerations	Notes
LVC MOS 1.8V	SE	1.8 V	2 – 16 mA	600MHz	-	-
LVC MOS 1.5V	SE	1.5 V	2 – 16 mA	600MHz	-	-
LVC MOS 1.2V	SE	1.2 V	2 – 12 mA	600MHz	-	-
SSTL 1.8V – I/II	SE/DIFF	1.8V	8.6 – 13.4 mA	1.6Gb/s	Controlled Source Impedance	DDR2
SSTL 1.5V	SE/DIFF	1.5V	8 – 16 mA	1.6Gb/s	Controlled Source Impedance	-
SSTL 1.2V – I/II	SE/DIFF	1.2V	-	1.2Gb/s	Controlled Source Impedance	-
HSTL 1.8V – I/II	SE/DIFF	1.8 V	8 – 16 mA	1.6Gb/s	Controlled Source Impedance	DDR2
HSTL 1.5V – I/II	SE/DIFF	1.5 V	8 – 16 mA	1.6Gb/s	Controlled Source Impedance	-
LVDS 1.8V	DIFF	1.8 V	-	800MHz	On-die differential termination	-

⁽¹⁾ 16 mA feature is being characterized

6.2.2 IO Complex Dedicated Pins

Clock pins

ULTRA300 has 2 dedicated preferred input clock pin per Complex IO bank.

Frequency limits on these 2 pins come from:

- PLL use: If the pin is directly connected to a PLL:
 - PLL frequency range is [10 – 50MHz] but can be extended using GCK input divider
 - If PLL is bypassed, WFG input range has the same limits as low-skew network
- Low-skew network:
 - It is not recommended to use a clock frequency higher than ~200MHz
 - Maximum frequency is design dependent. Only an STA can validate design frequency
 - Above 200MHz, duty cycle may be impacted

SpaceWire PHY

Using SpaceWire PHY dedicated pins (SPW0 and SPW1),

Table 17: IO Complex SpaceWire PHY performance [tab 3 - chap 6]

Protocol	Standard	Speed MHz or Mb/s	Special considerations
SpaceWire	LVDS 1.8V	400Mbps	Hardware SpaceWire PHY can be used only using dedicated SpaceWire pins: (SPW0 and SPW1)

6.3 Direct IO Banks

6.3.1 IO Direct Standard List

Table 18: IO Direct Standard List [tab 4 - chap 6]

Standard	Type	Supply	Drive	Speed MHz or Mb/s	Special considerations	Notes
LVC MOS 3.3V	SE	3.3 V	2 – 16 mA ⁽²⁾	200MHz ⁽¹⁾	-	-
LVC MOS 2.5V	SE	2.5 V	2 – 16 mA ⁽²⁾	200MHz ⁽¹⁾	-	-
LVC MOS 1.8V	SE	1.8 V	2 – 16 mA ⁽²⁾	200MHz ⁽¹⁾	-	-

⁽¹⁾ Maximum speed strongly depends on the driven output load

⁽²⁾ 16mA feature is being characterized

6.4 Analog-to-Digital Converters (ADC)

Table 19: ADC performances [tab 5 - chap 6]

Symbol	Parameter	Min	Typ	Max	Unit	Notes
VIN	Input voltage	0		1.6	V	The input absolute max value that can be temporarily applied without damage is 1.98V
fS	Sample rate		0.5	5	MSPS	
fIN	Input frequency	0	-	40	KHz	The input frequency should be limited at most to the Nyquist rate of the used sampling rate
SNDR	Signal-to-Noise Distortion Ratio (SNDR)		74		dB	

Symbol	Parameter	Min	Typ	Max	Unit	Notes
INL	Integral non-linearity			+/- 2.5	LSB	INL value is guaranteed for input range of 200mV – 1.4V
DNL	Differential non-linearity			+/- 0.1	LSB	

Full characterization of the ADC block is currently ongoing. ADC specifications and performance may be revised in a future version of this document.

6.5 Digital-to-Analog Converters (DAC)

Table 20: DAC performances [tab 6 - chap 6]

Symbol	Parameter	Min	Typ	Max	Unit	Notes
VOUT,CM	Output common mode voltage		0.9		V	
VOUT	Output voltage	0		1.6	V	
f _{out}	Output signal frequency	0	1	2	MHz	
f _S	Sample rate		1		MSPS	
INL	Integral non-linearity			+/- 2	LSB	INL value is guaranteed for input range: • 70mV – 1.65V (external +1.8V ref) • 70mV – 1.45V (Internal +1.6V ref)
DNL	Differential non-linearity			+/- 1	LSB	
t _{ST}	Settling time		50		ns	
CL	Output charge capacitance		50		pF	

Full characterization of the DAC block is currently ongoing. DAC specifications and performance may be revised in a future version of this document.

6.6 HSSL Transceivers

HSSL is currently under characterization. HSSL performances will be shared in a later update of the Datasheet.

7. Device interface

7.1 Introduction

The programmable logic of ULTRA300 FPGA is accessible through several dedicated IO banks available for user applications.

This chapter provides an overview of ULTRA300 IO organizations and features.

A dedicated chapter in ULTRA300 User Guide details IO bank features. Please refer to the ULTRA300 Documents chapter for the corresponding reference.

7.2 Device IO banks

ULTRA300 IO are organized in banks. ULTRA300 has 5 different types of IO banks:

- Service IO bank: Bank dedicated to ULTRA300 Service (boot, Power Status, etc...) and partially available as user IO
- Complex IO banks: Banks dedicated to high speed user IO with advanced settings
- Direct IO banks: Banks dedicated to general purpose user IO with simple settings
- HSSL IO banks: Banks dedicated to high Speed Serial Links (Transceivers)
- ADC and DAC IO banks: Banks dedicated to ADC and DAC interfaces

7.2.1 Service IO bank

ULTRA300 has 1 service bank. Service IO bank is dedicated to ULTRA300 service functions: boot, status and has some dual-purpose IO available as user IO after the boot sequence.

Service IO pads are the same type as Direct IO banks. Service bank contains up to 57 IO pads.

Service IO bank operates with a 3.3V power supply.

The service bank offers a wide range of configuration interfaces and modes. Configuration pins include:

- Configuration select pins
- JTAG
- 3xQSPI controller: SPI0, SPI1 and SPI2
 - Compatible with single, dual and QUAD SPI modes
 - Independent or TMR mode
- SpaceWire
- 16b Parallel

Configuration interfaces are located on specific pins. Several pins are shared between the different configuration interfaces. A dedicated User Guide detail the configuration process and pins. Please refer to the ULTRA300 Documents chapter for the corresponding reference.

The service bank also contains several pins to control and monitor ULTRA300 status:

- FPGA ID
- Reset inputs
- Power OK
- Status flags

Service IO banks settings

ULTRA300 Service IO banks pads are the same type as Direct Bank pads but with a fixed configuration:

- Drive Strength: 16 mA⁽¹⁾ for all pads
- IO Direction: pads can be either IN, OUT or INOUT, refer to [Service IO Bank pads list \[tab 1 - chap 7\]](#)
- IO Weak Termination: pads can have either a weak pull-up, a weak pull-down and no termination, refer to [Service IO Bank pads list \[tab 1 - chap 7\]](#)

⁽¹⁾ 16mA feature is being characterized

Service IO banks pins

Service IO Bank pads list [tab 1 - chap 7] details ULTRA300 Service bank pins. Some pins are not connected in 484 pins package. Please check Package FBGA484 chapter for detailed information.

Table 21: Service IO Bank pads list [tab 1 - chap 7]						
PAD	Direction	Type	Standard	Termination	Voltage (V)	Description
CLK_OSC ⁽¹⁾	OUT	Direct	LVC MOS	None	3.3	Bitstream Manager internal Clock Oscillator
CLK_BSM	IN	Direct	LVC MOS	None	3.3	Parallel16 interface clock when PAR16 configuration mode is used
RST_SOFT_N	IN	Direct	LVC MOS	Weak pull-up	3.3	Additional reset signal to be used in combination of RST_HARD_N
RST_HARD_N	IN	Direct	LVC MOS	Weak pull-up	3.3	Main reset signal. Reset Bitstream manager and Fabric
POK ⁽¹⁾	OUT	Direct	LVC MOS	None	3.3	Power OK
MODE[3:0]	IN	Direct	LVC MOS	Weak pull-down	3.3	Configuration Mode Selector
ID[3:2] ⁽²⁾	IN	Direct	LVC MOS	Weak pull-down	3.3	FPGA identification (4 pins)
ID[1:0]						
READY	OUT	Direct	LVC MOS	None	3.3	Ready flag
TRIGGER0	OUT	Direct	LVC MOS	None	3.3	Error flag
TRIGGER1	OUT	Direct	LVC MOS	None	3.3	Trigger flag
TCK	IN	Direct	LVC MOS	None	3.3	JTAG clock
TRST	IN	Direct	LVC MOS	Weak pull-up	3.3	JTAG reset
TMS	IN	Direct	LVC MOS	Weak pull-up	3.3	JTAG mode
TDI	IN	Direct	LVC MOS	Weak pull-up	3.3	JTAG input
TDO	OUTZ	Direct	LVC MOS	Weak pull-down	3.3	JTAG output
CS	INOUT	Direct	LVC MOS	Weak pull-down	3.3	Shared IO used for configuration (SPI1 and parallel 16)
TYPE[1:0]	INOUT	Direct	LVC MOS	Weak pull-down	3.3	Shared IOs used for configuration (SPI2 and parallel 16)
DATA_OE	OUT	Direct	LVC MOS	Weak pull-down	3.3	Shared IO used for configuration (SPI1 and parallel 16)

PAD	Direction	Type	Standard	Termination	Voltage (V)	Description
[D15:D0]	INOUT	Direct	LVC MOS	Weak pull-down	3.3	Shared IOs used for configuration (SPI0, SPI2 and parallel 16)
USER[10:0]	INOUT	Direct	LVC MOS	Weak pull-down	3.3	Shared IOs used for configuration (SPI1 and SPI2) or user application
DIN ⁽²⁾	IN	Direct	LVC MOS	None	3.3	SpaceWire Data In
DOUT ⁽¹⁾	OUTZ	Direct	LVC MOS	None	3.3	SpaceWire Data Out
SIN ⁽²⁾	IN	Direct	LVC MOS	None	3.3	SpaceWire Strobe In
SOUT ⁽¹⁾	OUTZ	Direct	LVC MOS	None	3.3	SpaceWire Strobe Out
TEST_MODE ⁽²⁾	IN	Direct	LVC MOS	Weak pull-down	3.3	Reserved

(1) Not connected for 484 pins package

(2) Connected to ground for 484 pins package

7.2.2 Complex IO banks – High Speed IO (HSIO)

ULTRA300 provides up to 10 Complex IO banks. Complex IO banks are dedicated to user IO.

Each Complex IO bank is made of 34 bidirectional IO pads and is organized in 3 sub-banks:

- Sub-bank0
- Sub-bank1
- Sub-bank2

Complex IO pads support single-ended and differential IO standards (34 single or 17 differential IO pads per bank).

Complex IO banks can operate with 1.2V, 1.5V or 1.8V power supplies.

Complex IO pads are compatible with complex interfaces such as SpaceWire and DDR2.

Each complex IO bank provides:

- 2 dedicated input differential Clock pins
 - Preferred pins for clock input
 - These pins enhance clocking performances using dedicated lowskew network connections
- 2 x RX/TX SpaceWire Interface: SPW0 and SPW1
 - Hardware optimized PHY for max performances
 - SpaceWire speed will be limited unless using these dedicated pins
 - SPW0 is located in sub-bank 0
 - SPW1 is located in sub-bank 2
- Dedicated DDR interfaces:
 - Specific rules must be followed to interface a DDR memory. Pin selection rules are detailed in dedicated chapter [“DDR Support in Complex IO bank”](#).

Complex IO banks list

ULTRA300 has up to 10 Complex IO banks:

Table 22: Complex IO bank list [tab 2 - chap 7]

Bank ID	Bank Type	Comment
IOB03	Complex	Only available in 1152 pins package
IOB04	Complex	-
IOB05	Complex	Only available in 1152 pins package
IOB06	Complex	Only available in 1152 pins package
IOB09	Complex	Only available in 1152 pins package
IOB10	Complex	Only available in 1152 pins package
IOB11	Complex	Only available in 1152 pins package
IOB15	Complex	-
IOB16	Complex	-
IOB17	Complex	-

Refer to “Packages chapter” for more detail about each package.

Complex IO pads

A Complex IO pad is a programmable IO pad providing an IO Buffer supporting single-ended and differential standards along with various features. Two adjacent IO Buffers can be paired to form a differential buffer. A programmable resistor network provides on-chip termination connected to an external voltage VTO, or a differential resistive termination between two paired IOBs.

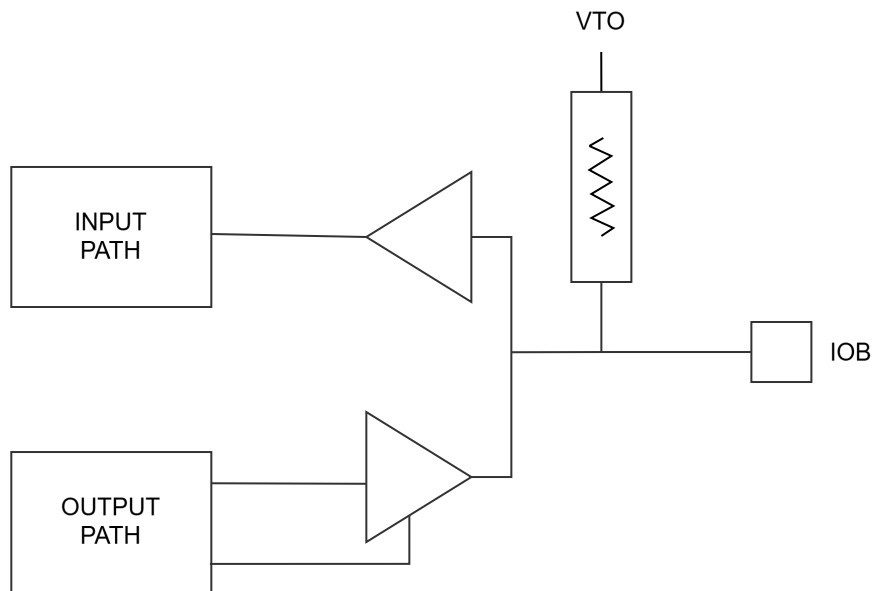


Figure 8: Complex IOB with single-ended termination [fig 1 - chap 7]

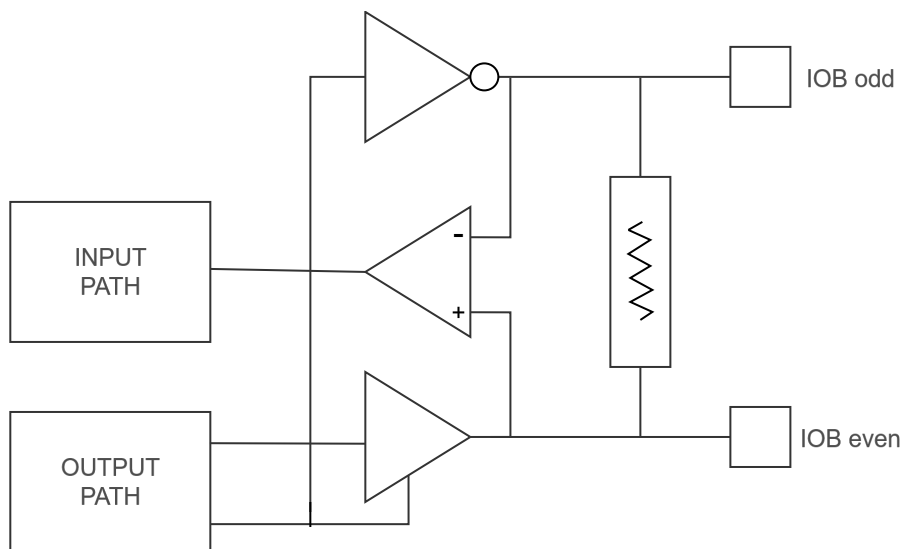


Figure 9: Differential IOB with a differential termination [fig 2 - chap 7]

Complex IOs are assembled in pairs as P/N. Each pair may be used as either a true differential signal or two unipolar signals. The differential signaling and differential termination options are configured separately; therefore, a differential pair can be programmed to have a single-ended or differential termination scheme.

A Complex IO Bank is composed of several Complex IOs and supply IOs forming a homogeneous structure sharing:

- A same IO power supply VDDIO
- A same resistive termination supply VTO
- A same On-Chip termination buffer switch supply VDD_SWITCH

A Complex IO Bank supports single-ended and differential IO standards allowing IO supplies of 1.2V, 1.5V or 1.8V and standard interfaces:

- DDR2 interface capability which is standard dependent
- Controlled impedance termination capabilities
- SpaceWire interface capability

Table 23: Complex IO pad features [tab 3 - chap 7]

IO Complex Features
Support multiple voltage for single-ended and differential standards (1.2V/1.5/1.8V)
Driver buffer with programmable drive strength and weak termination buffer (bus-hold)
Input buffer with internally generated programmable reference voltage
Support of differential signaling
On-die programmable termination (single-ended and differential)
Programmable delay lines for input and output data paths
Programmable pre-emphasis and slew rate control
ESD protection
Cold-spare protection
Optional single or double rate register

Complex IO banks pins

Complex IO Bank pads list [tab 4 - chap 7] details ULTRA300 Complex IO bank pins.

Table 24: Complex IO Bank pads list [tab 4 - chap 7]							
PADS name	ID	Sub Bank	Direction	Type	Standard	Voltage (V)	Description
IOB*_D00P_SWDO	0	0	INOUT	Complex	LVDS / LVCMOS / SSTL / HSTL	1.2 / 1.5 / 1.8	SPW0: Data_out_p or GPIO
IOB*_D00N_SWDO	1	0	INOUT	Complex	LVDS / LVCMOS / SSTL / HSTL	1.2 / 1.5 / 1.8	SPW0: Data_out_n or GPIO
IOB*_D01P_SWSO	2	0	INOUT	Complex	LVDS / LVCMOS / SSTL / HSTL	1.2 / 1.5 / 1.8	SPW0: Strobe_out_p or GPIO
IOB*_D01N_SWSO	3	0	INOUT	Complex	LVDS / LVCMOS / SSTL / HSTL	1.2 / 1.5 / 1.8	SPW0: Strobe_out_p or GPIO
IOB*_D02P_SWDI	4	0	INOUT	Complex	LVDS / LVCMOS / SSTL / HSTL	1.2 / 1.5 / 1.8	SPW0: Data_in_p or GPIO
IOB*_D02N_SWDI	5	0	INOUT	Complex	LVDS / LVCMOS / SSTL / HSTL	1.2 / 1.5 / 1.8	SPW0: Data_in_p or GPIO
IOB*_D03P_SWSI	6	0	INOUT	Complex	LVDS / LVCMOS / SSTL / HSTL	1.2 / 1.5 / 1.8	SPW0: Strobe_in_p or GPIO
IOB*_D03N_SWSI	7	0	INOUT	Complex	LVDS / LVCMOS / SSTL / HSTL	1.2 / 1.5 / 1.8	SPW0: Strobe_in_p or GPIO
IOB*_D04P	8	0	INOUT	Complex	LVDS / LVCMOS / SSTL / HSTL	1.2 / 1.5 / 1.8	GPIO
IOB*_D04N	9	0	INOUT	Complex	LVDS / LVCMOS / SSTL / HSTL	1.2 / 1.5 / 1.8	GPIO
IOB*_D05P	10	0	INOUT	Complex	LVDS / LVCMOS / SSTL / HSTL	1.2 / 1.5 / 1.8	GPIO
IOB*_D05N	11	1	INOUT	Complex	LVDS / LVCMOS / SSTL / HSTL	1.2 / 1.5 / 1.8	GPIO
IOB*_D06P	12	1	INOUT	Complex	LVDS / LVCMOS / SSTL / HSTL	1.2 / 1.5 / 1.8	GPIO
IOB*_D06N	13	1	INOUT	Complex	LVDS / LVCMOS / SSTL / HSTL	1.2 / 1.5 / 1.8	GPIO

PADS name	ID	Sub Bank	Direction	Type	Standard	Voltage (V)	Description
IOB*_D07P	14	1	INOUT	Complex	LVDS / LVCMOS / SSTL / HSTL	1.2 / 1.5 / 1.8	GPIO
IOB*_D07N	15	1	INOUT	Complex	LVDS / LVCMOS / SSTL / HSTL	1.2 / 1.5 / 1.8	GPIO
IOB*_D08P_CLK	16	1	INOUT	Complex	LVDS / LVCMOS / SSTL / HSTL	1.2 / 1.5 / 1.8	Dedicated input clock pin P or GPIO
IOB*_D08N	17	1	INOUT	Complex	LVDS / LVCMOS / SSTL / HSTL	1.2 / 1.5 / 1.8	Dedicated input differential clock pin N or GPIO
IOB*_D09P_CLK	18	1	INOUT	Complex	LVDS / LVCMOS / SSTL / HSTL	1.2 / 1.5 / 1.8	Dedicated input clock pin P or GPIO
IOB*_D09N	19	1	INOUT	Complex	LVDS / LVCMOS / SSTL / HSTL	1.2 / 1.5 / 1.8	Dedicated input differential clock pin N or GPIO
IOB*_D10P	20	1	INOUT	Complex	LVDS / LVCMOS / SSTL / HSTL	1.2 / 1.5 / 1.8	GPIO
IOB*_D10N	21	1	INOUT	Complex	LVDS / LVCMOS / SSTL / HSTL	1.2 / 1.5 / 1.8	GPIO
IOB*_D11P	22	2	INOUT	Complex	LVDS / LVCMOS / SSTL / HSTL	1.2 / 1.5 / 1.8	GPIO
IOB*_D11N	23	2	INOUT	Complex	LVDS / LVCMOS / SSTL / HSTL	1.2 / 1.5 / 1.8	GPIO
IOB*_D12P	24	2	INOUT	Complex	LVDS / LVCMOS / SSTL / HSTL	1.2 / 1.5 / 1.8	GPIO
IOB*_D12N	25	2	INOUT	Complex	LVDS / LVCMOS / SSTL / HSTL	1.2 / 1.5 / 1.8	GPIO
IOB*_D13P_SWDO	26	2	INOUT	Complex	LVDS / LVCMOS / SSTL / HSTL	1.2 / 1.5 / 1.8	SPW1: Data_out_p or GPIO
IOB*_D13N_SWDO	27	2	INOUT	Complex	LVDS / LVCMOS / SSTL / HSTL	1.2 / 1.5 / 1.8	SPW1: Data_out_n or GPIO
IOB*_D14P_SWSO	28	2	INOUT	Complex	LVDS / LVCMOS / SSTL / HSTL	1.2 / 1.5 / 1.8	SPW1: Strobe_out_p or GPIO
IOB*_D14N_SWSO	29	2	INOUT	Complex	LVDS / LVCMOS / SSTL / HSTL	1.2 / 1.5 / 1.8	SPW1: Strobe_out_p or GPIO

PADS name	ID	Sub Bank	Direction	Type	Standard	Voltage (V)	Description
IOB*_D15P_SWDI	30	2	INOUT	Complex	LVDS / LVCMOS / SSTL / HSTL	1.2 / 1.5 / 1.8	SPW1: Data_in_p or GPIO
IOB*_D15N_SWDI	31	2	INOUT	Complex	LVDS / LVCMOS / SSTL / HSTL	1.2 / 1.5 / 1.8	SPW1: Data_in_p or GPIO
IOB*_D16P_SWSI	32	2	INOUT	Complex	LVDS / LVCMOS / SSTL / HSTL	1.2 / 1.5 / 1.8	SPW1: Strobe_in_p or GPIO
IOB*_D16N_SWSI	33	2	INOUT	Complex	LVDS / LVCMOS / SSTL / HSTL	1.2 / 1.5 / 1.8	SPW1: Strobe_in_p or GPIO

Some Complex IO banks are not connected in 484 pins package. Please check Package FBGA484 chapter for detailed information.

Complex IO pad On-chip resistive termination

The programmable on-chip resistive termination network is controlled through 4-bits calibration control signal. This 4-bit value is assigned to the IOs by configuration for each IO requiring internal input impedance adaptation.

Possible termination schemes [fig 3 - chap 7] presents two possible termination schemes. In both configurations, the switches connecting termination resistance (Rterm) to termination point (VTO power rail in single-ended mode, or common point in differential mode) are controlled by command signals in VDDS power domain.

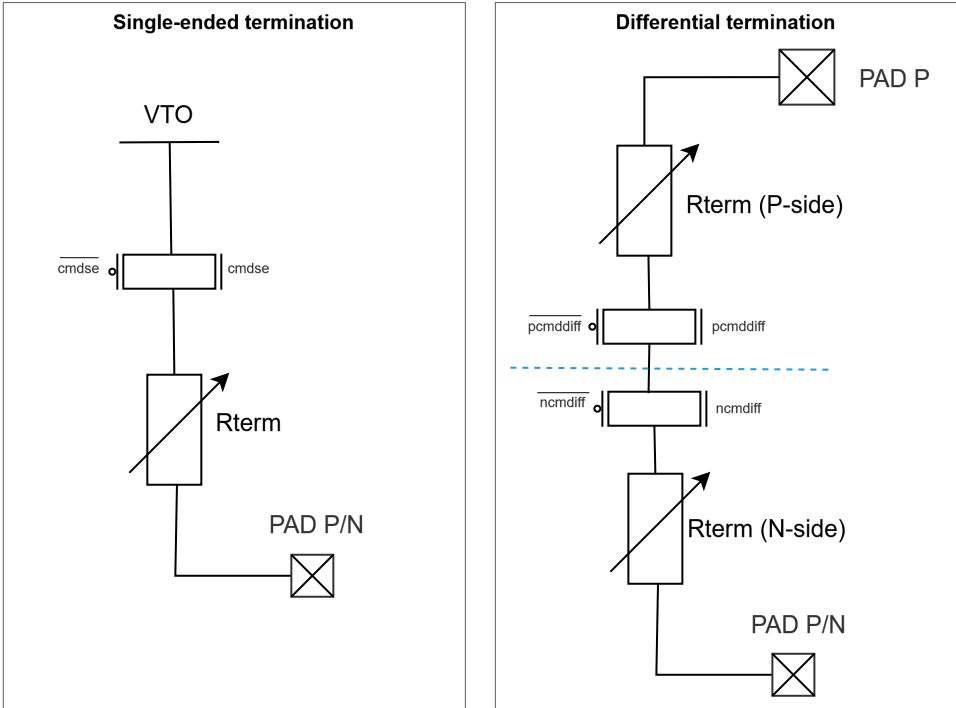


Figure 10: Possible termination schemes [fig 3 - chap 7]

In single-ended mode, dedicated IO Bank VTO pins must be connected to an external VTO voltage.

In case where the on-die termination, feature is configured as differential:

- VTO power rail can be left connected, unconnected or grounded.
- VDDS should be connected to 1V8

In case where the on-die termination, feature is not used,

- VTO power rail can be left connected, unconnected or grounded.
- If VTO is not floating, VDDS should be connected
- If VTO floating VDDS can be connected, unconnected or grounded.

If termination feature is enabled, VDDS should be powered.

VDDS value depends both on the termination mode and VTO connection type. [Termination Voltage conditions \[tab 5 - chap 7\]](#) details VDDS power rail voltage values depending on these conditions.

Table 25: Termination Voltage conditions [tab 5 - chap 7]

Mode	VTO	VDDS
Single-Ended Termination	Powered ⁽¹⁾⁽²⁾	1.8V
Differential Termination	Powered ⁽¹⁾⁽²⁾ or Grounded or Floating (not used)	1.8V
Termination not used	Floating	1.8V or Grounded or Floating
Termination not used	Grounded	1.8V

⁽¹⁾ VTO power rail voltage depends on the used IO standard.

⁽²⁾ The “VTO Powered” condition includes cases where the termination buffer is not used in the design, but the VTO rail is still powered on the PCB.

Impedance adaptation resistor values for VDDIO=1.2V [fig 4 - chap 7] to Impedance adaptation resistor values for VDDIO=1.8V [fig 6 - chap 7] charts present on-die termination resistor values covering PVT variations. These Charts values are indicative and given as examples to understand the calibration codes settings and PVT effects.

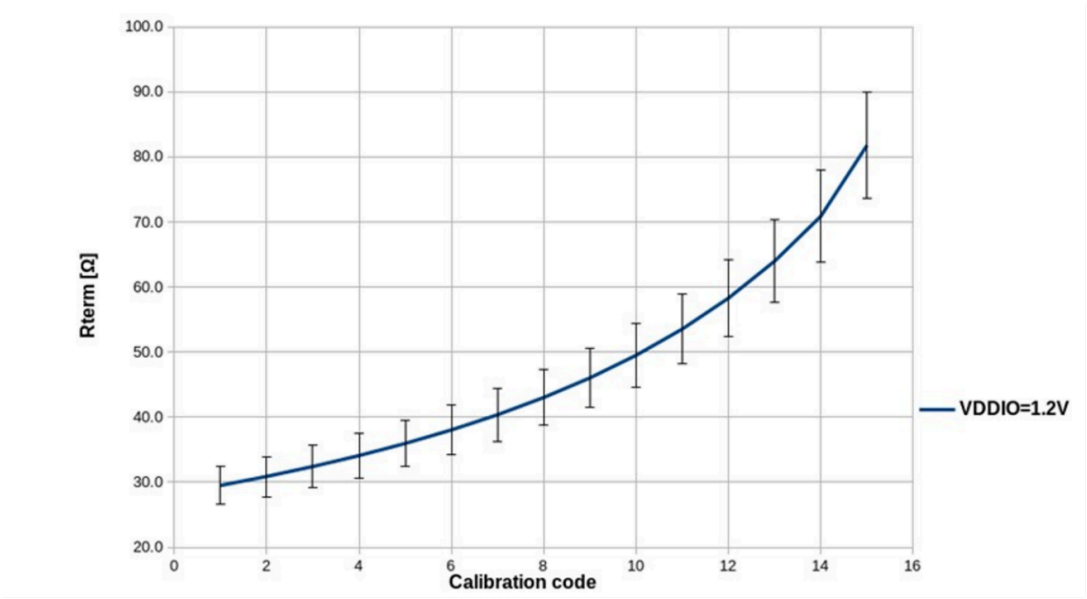


Figure 11: Impedance adaptation resistor values for VDDIO=1.2V [fig 4 - chap 7]

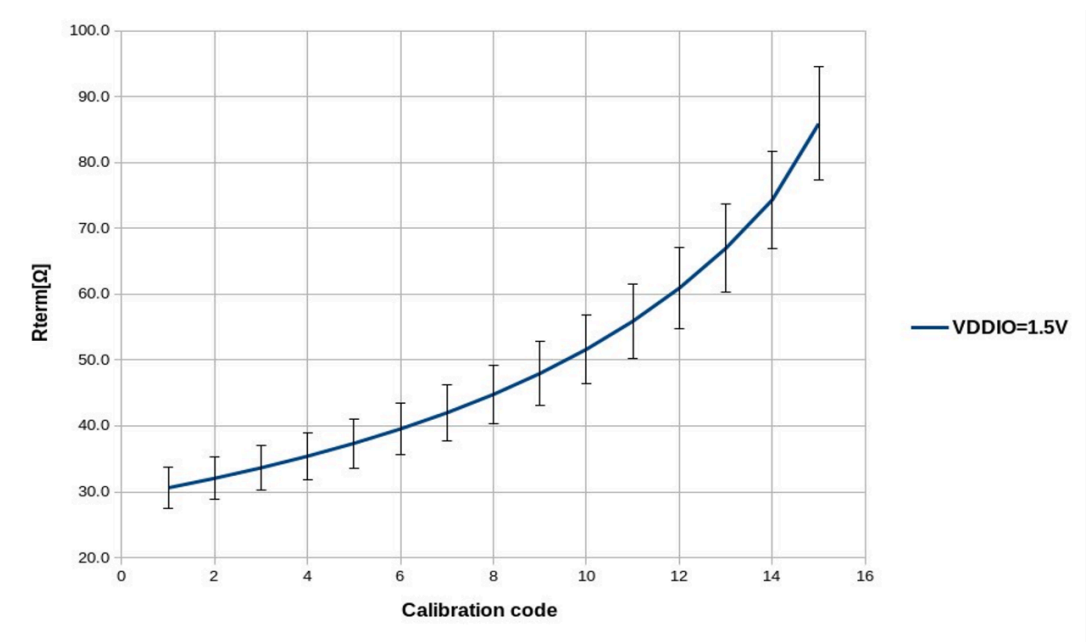


Figure 12: Impedance adaptation resistor values for VDDIO=1.5V [fig 5 - chap 7]

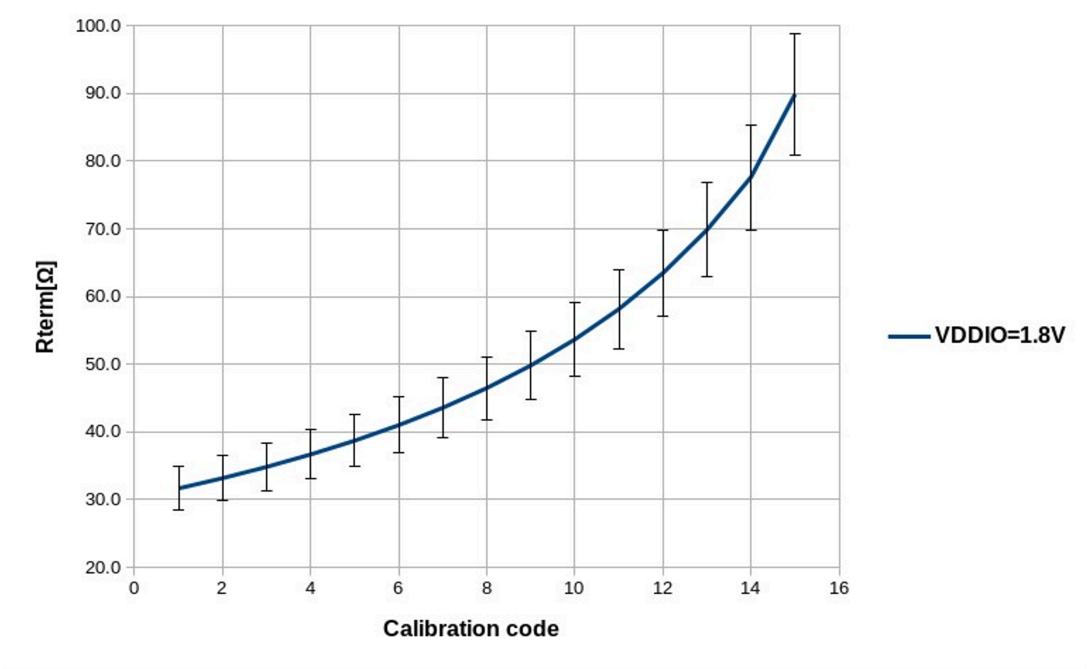


Figure 13: Impedance adaptation resistor values for VDDIO=1.8V [fig 6 - chap 7]

Complex IO Pad Delay lines

Each Complex IO Buffer has programmable delay lines on their input and output paths and can be combined with a register, shift registers and a CDC function present in the IO Bank logic to provide various types of data interface and clocking modes with the fabric. Overall, there are four delay lines; two delay lines are dedicated for output path addressing data and output enable signals, one delay line each for input data signal and termination control. The programmable delay line is used to align incoming or outgoing data with the external clocking environment and can be bypassed.

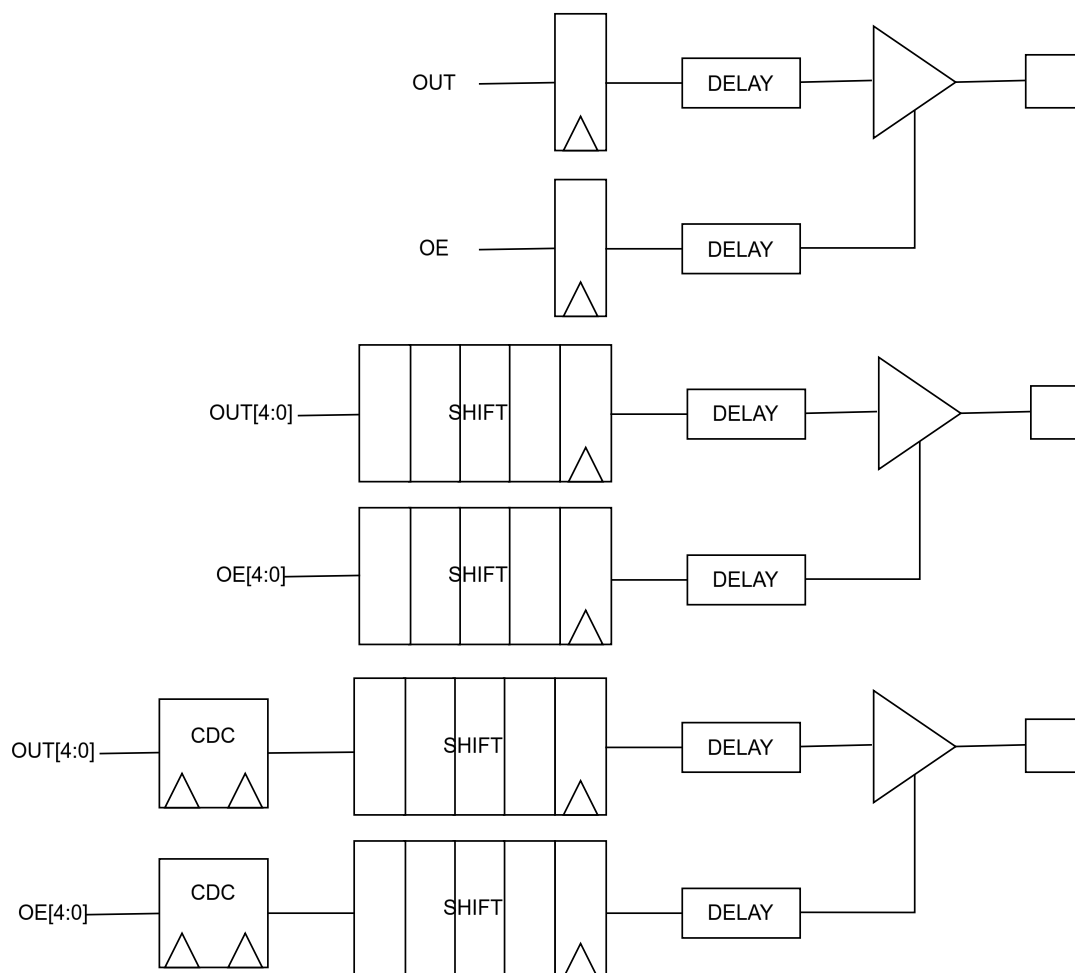


Figure 14: Output Path possible configurations with delay line [fig 7 - chap 7]

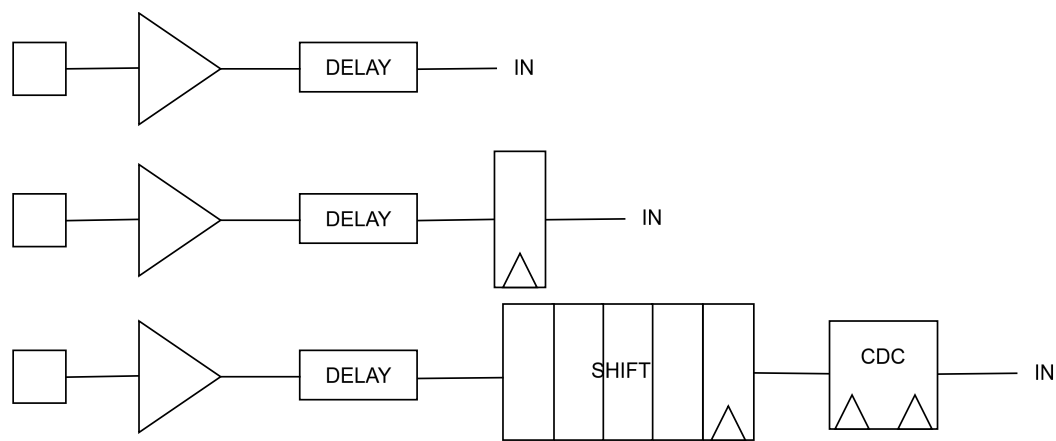


Figure 15: Input Path possible configuration with delay line [fig 8 - chap 7]

The delay lines are PVT compensated. Delay step calibration is through 5 bits (0-31) and delay tap value selection is through 6 bits (0-63).

Table 26: Complex IO Delay line parameters [tab 6 - chap 7]	
Parameter	Value
IO delay tap	100ps ± 11%
Intra Pair (P/N) delta delay	± 4% of IO delay tap
Delay Line offset	[110 – 120] ps
Delay Line total delay	$Delay = Offset + (TapVal + 1) * Tap$

The delay line can be bypassed.

Complex IO Pad Weak-Termination

Weak termination buffer features bus-hold (keeper) structure. Weak pull-up resistor, weak pull-down resistor and bus-hold function are accessed through configuration. The selected configuration is exclusive meaning that other possible functions can be activated at the same time in the same IO. Each IO in each bank can be configured separately.

Weak termination buffer supports cold-sparing mode (no current path from PAD to chip when the chip is off).

To enable bus-hold mode, both output buffer and input buffer must be enabled.

Table 27: IO Comple Weak termination buffer bus-hold (keeper) characteristics [tab 7 - chap 7]

Parameter	Symbol	VDDIO [V]					
		1.2		1.5		1.8	
		min	max	min	max	min	max
Bus-hold low, sustaining current [uA]	IBHL	6.7	-	11.7	-	17.5	-
Bus-hold high, sustaining current [uA]	IBHH	-5.6	-	-12.6	-	-20.2	-
Bus-hold low, overdrive current [uA]	IBHLO	-	24.0	-	40.5	-	56.5
Bus-hold high, overdrive current [uA]	IBHLO	-	-15.5	-	-33.5	-	-53.4

Table 28: IO Complex Weak Termination Rpullup and Rpulldown resistance
minimum and maximum values during product lifetime [tab 8 - chap 7]

Parameter	VDDIO [V]					
	1.2		1.5		1.8	
	min	max	min	max	min	max
Rpulldown [kΩ]	17.5	45.0	13.2	32.0	11.3	25.7
Rpullup [kΩ]	21.5	53.5	15.4	29.8	12.7	22.3

Complex IO Pad Pre-Emphasis and Slew-rate control

IO output buffer can provide three different modes for output signal transition times: Medium (Normal), Fast and Slow.

In Fast mode, pre-emphasis function (edge boost) allows boosting output transition rate. The boost rate depends on logic signal edge rates, VDDIO voltage value, the selected drive current strength and the driven output load.

In Slow mode, the output buffer transition rate is reduced. This reduction rate depends on the output buffer input signal transition times, the selected drive current strength and the driven output load.

DDR Support in Complex IO bank

11 IOB can be combined to form a DDR physical byte lane interface ([DDR Physical Interface\[fig 9 - chap 7\]](#)) with 9 DQ (8 data, 1 data mask) and 2 DQS (differential strobe). The DDR INTERFACE will make data and clock alignment, and data serialization.

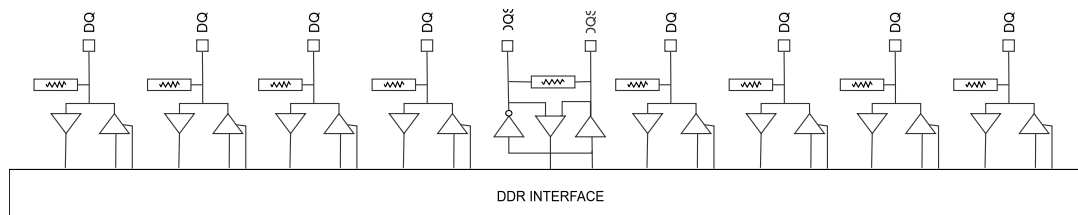


Figure 16: [DDR Physical Interface\[fig 9 - chap 7\]](#)

Bytelanes must be gathered in the same sub-bank (3 sub-banks per Complex IO bank), as detailed on [DDR complex sub-banks \(IO Path = IO ID\) \[fig 10 - chap 7\]](#). [Complex IO Bank pads list \[tab 4 - chap 7\]](#) help to make the connection between “IO PATH” and “Pad name”.

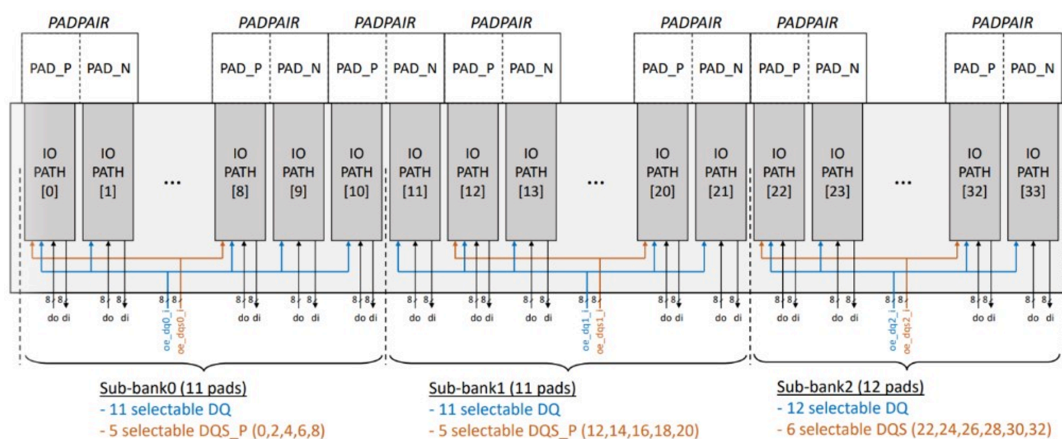


Figure 17: [DDR complex sub-banks \(IO Path = IO ID\) \[fig 10 - chap 7\]](#)

DDR interface pin selection rules are

- Each sub-bank can host a DDR bytelane
 - 1 bytelane = 11 pins: 1DQS differential pair, 8 DQ and 1 DM
 - Bytelane pins must be gathered in the same sub-bank
 - There is no preferred pin for DQ, DQS or DM pins in a sub-bank
- Clocks, commands and control pins may be located anywhere in Complex IO banks
 - All these pins MUST be clocked using clocks from the same CKG as associated bytelanes
 - These pins may or may not be located in the same Complex IO banks as associated bytelanes

SpaceWire Support in Complex IO bank

4 IOB and dedicated TX and RX modules can be combined to implement a physical media access layer of a SpaceWire interface ([SpaceWire Physical Interface \[fig 11 - chap 7\]](#)).

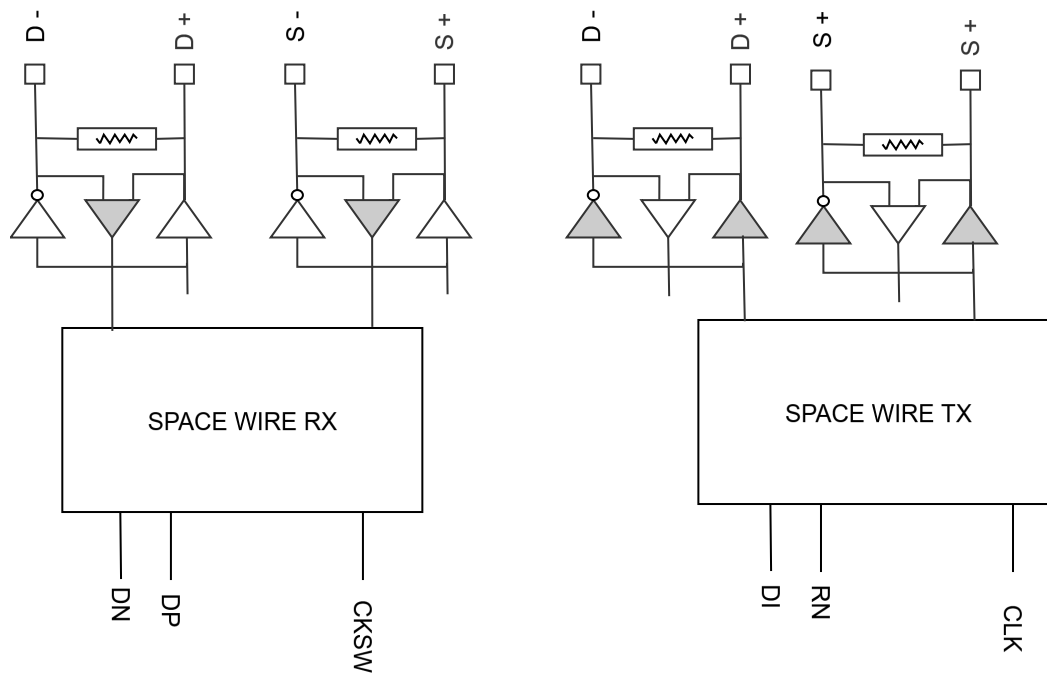


Figure 18: SpaceWire Physical Interface [fig 11 - chap 7]

The SpaceWire TX module receives the transmit clock (CLK), a reset input (RN) and the parallel data input (DI) and generates the Data Strobe and the serial data on LVDS outputs. The RX module receives the Data Strobe LVDS inputs and generates the data clock (CKSW), the positive edge data (DP) and the negative edge data (DN). The SpaceWire RX and TX modules provided in the Complex IO Banks don't provide serializer/deserializer features.

NanoXplore provides a SpaceWire Bank IP that can be instantiated to configure SpaceWire protocol with full capacities above a Complex IO Bank. The overview of this SpaceWire IP is represented in [SpaceWire IP overview\[fig 12 - chap 7\]](#).

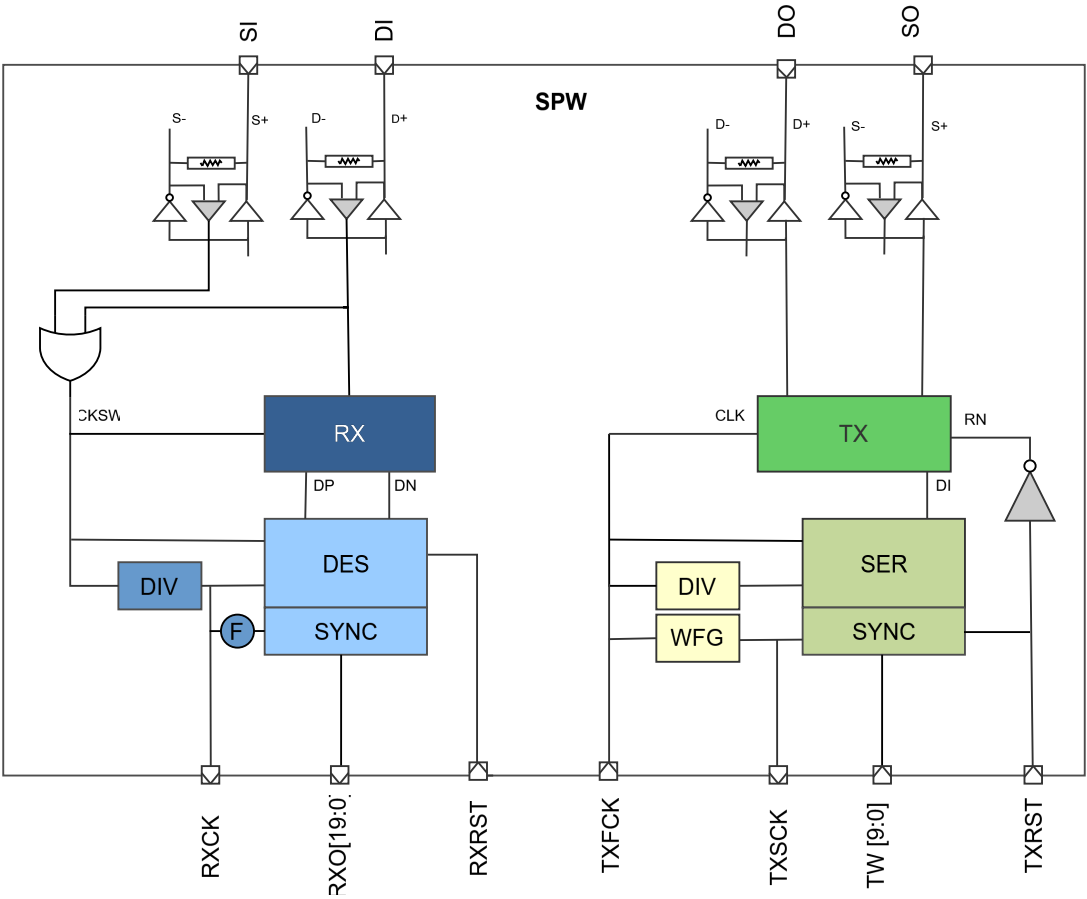


Figure 19: SpaceWire IP overview [fig 12 - chap 7]

Each Complex IO banks host 2 RX/TX SpaceWire interfaces: SPW0 and SPW1. Dedicated pins must be used to access these SpaceWire interface. They are listed in [Complex IO Bank pads list \[tab 4 - chap 7\]](#).

7.2.3 Direct IO banks – General Purpose IO (GPIO)

ULTRA300 provides up to 7 Direct IO banks. Direct IO banks are dedicated to user IO.

Direct IO banks are made of maximum 24 bidirectional IO pads (“large Direct IO bank”) or 16 bidirectional IO pads (“small Direct IO bank”).

Direct IO pads support only single-ended IO standards.

Direct IO banks can operate with 1.8V, 2.5V or 3.3V power supplies.

All Direct I/O pads are identical. Unlike Complex pads, they do not provide any dedicated access to the clock network on specific pins.

Direct IO banks list

ULTRA300 has up to 7 Direct IO banks:

Table 29: Direct IO bank list [tab 9 - chap 7]

Bank ID	Bank Type	Comment
IOB01	Direct	Only available in 1152 pins package
IOB02	Direct	Only available in 1152 pins package
IOB07	Direct	-
IOB08	Direct	Limited pin number in 484 pins package
IOB12	Direct	Bank is Shared with ADC/DAC in 484 pins package
IOB13	Direct	Bank is Shared with ADC/DAC in 484 pins package
IOB14	Direct	Bank is Shared with ADC/DAC in 484 pins package

Check Packages chapter for more detail about each package.

Direct IO pad

A Direct IO pad is a programmable IO pad providing a single-ended IO Buffer and several features.

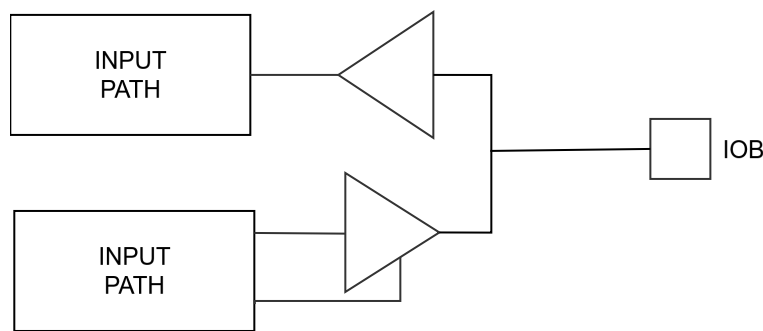


Figure 20: Single Ended IOB in Direct IO pad [fig 13 - chap 7]

Table 30: Direct IO pad features [tab 10 - chap 7]

IO Direct Features
Support multiple voltage for single-ended standards (1.8V/2.5V/3.3)
Driver buffer with programmable drive strength and weak termination buffer (bus-hold)
Input buffer
ESD protection
Cold-spare protection
Optional single rate register

Direct IO banks pins

ULTRA300 FPGA has 2 different types of Direct IO banks:

- Large banks: 24 pins
- Small banks: 16 pins
- 484 pins package has 1 bank limited to 15 pins

Direct IO Bank pads list [tab 11 - chap 7] details ULTRA300 Direct bank pins for a “large IO bank”. (Bank 13 as only pads D00..D15).

Table 31: Direct IO Bank pads list [tab 11 - chap 7]

PADS	Direction	Type	Standard	Voltage (V)	Description
IOB*_D00	INOUT	Direct	LVC MOS	1.8 / 2.5 / 3.3	GPIO
IOB*_D01	INOUT	Direct	LVC MOS	1.8 / 2.5 / 3.3	GPIO
IOB*_D02	INOUT	Direct	LVC MOS	1.8 / 2.5 / 3.3	GPIO
IOB*_D03	INOUT	Direct	LVC MOS	1.8 / 2.5 / 3.3	GPIO
IOB*_D04	INOUT	Direct	LVC MOS	1.8 / 2.5 / 3.3	GPIO
IOB*_D05	INOUT	Direct	LVC MOS	1.8 / 2.5 / 3.3	GPIO
IOB*_D06	INOUT	Direct	LVC MOS	1.8 / 2.5 / 3.3	GPIO
IOB*_D07	INOUT	Direct	LVC MOS	1.8 / 2.5 / 3.3	GPIO
IOB*_D08	INOUT	Direct	LVC MOS	1.8 / 2.5 / 3.3	GPIO
IOB*_D09	INOUT	Direct	LVC MOS	1.8 / 2.5 / 3.3	GPIO
IOB*_D10	INOUT	Direct	LVC MOS	1.8 / 2.5 / 3.3	GPIO
IOB*_D11	INOUT	Direct	LVC MOS	1.8 / 2.5 / 3.3	GPIO
IOB*_D12	INOUT	Direct	LVC MOS	1.8 / 2.5 / 3.3	GPIO
IOB*_D13	INOUT	Direct	LVC MOS	1.8 / 2.5 / 3.3	GPIO
IOB*_D14	INOUT	Direct	LVC MOS	1.8 / 2.5 / 3.3	GPIO
IOB*_D15	INOUT	Direct	LVC MOS	1.8 / 2.5 / 3.3	GPIO
IOB*_D16	INOUT	Direct	LVC MOS	1.8 / 2.5 / 3.3	GPIO
IOB*_D17	INOUT	Direct	LVC MOS	1.8 / 2.5 / 3.3	GPIO
IOB*_D18	INOUT	Direct	LVC MOS	1.8 / 2.5 / 3.3	GPIO
IOB*_D19	INOUT	Direct	LVC MOS	1.8 / 2.5 / 3.3	GPIO
IOB*_D20	INOUT	Direct	LVC MOS	1.8 / 2.5 / 3.3	GPIO
IOB*_D21	INOUT	Direct	LVC MOS	1.8 / 2.5 / 3.3	GPIO
IOB*_D22	INOUT	Direct	LVC MOS	1.8 / 2.5 / 3.3	GPIO
IOB*_D23	INOUT	Direct	LVC MOS	1.8 / 2.5 / 3.3	GPIO

Some banks/pads are not connected in 484 pins package. Please check Package FBGA484 chapter for detailed information.

Direct IO Weak termination

Weak termination buffer features bus-hold (keeper) structure. Weak pull-up resistor, weak pull-down resistor and bus-hold function are accessed through configuration. The selected configuration is exclusive meaning that other possible functions can be activated at the same time in the same IO. Each IO in a given IO Bank can be configured separately.

Weak termination buffer supports cold-sparing mode (no current path from PAD to chip when the chip is off).

To enable bus-hold mode, both output buffer and input buffer must be enabled.

Table 32: IO Direct weak termination buffer bus-hold (keeper) characteristics [tab 12 - chap 7]

Parameter	Symbol	VDDIO [V]					
		1.8		2.5		3.3	
		min	max	min	max	min	max
Bus-hold low, sustaining current [uA]	IBHL	20.0	-	25.3		29.6	-
Bus-hold high, sustaining current [uA]	IBHH	-23.3	-	-28.5		-31.7	-
Bus-hold low, overdrive current [uA]	IBHLO	-	72.6	-	76.7	-	76.2
Bus-hold high, overdrive current [uA]	IBHLO	-	-66.5	-	-67.9	-	-65.6

Table 33: IO Direct weak termination buffer Rpullup and Rpulldown resistance minimum and maximum values during product lifetime [tab 13 - chap 7]

	VDDIO [V]					
	1.8		2.5		3.3	
	min	max	min	max	min	max
Rpulldown [kΩ]	9.4	22.5	10.2	24.7	11.5	27.8
Rpullup [kΩ]	11.1	19.3	12.3	21.9	14.3	26.1

7.2.4 HSSL IO banks - Transceivers

ULTRA300 provides 4 HSSL IO banks for a total of 16 high speed SERDES links.

Each HSSL IO bank is composed of 4TX/RX lanes, 1 PLL and 1 calibration. Each lane has its dedicated PMA and PCS functionalities.

Each HSSL bank has its own dedicated power supplies.

Note:

HSSL feature is currently under characterization. HSSL recommendations will shared in a later update of the Datasheet

HSSL IO banks list

ULTRA300 has up to 4 HSSL IO banks:

Table 34: HSSL IO bank list [tab 14 - chap 7]

Bank ID	Bank Type	Comment
HSSL00	HSSL	Only available in 1152 pins package
HSSL01	Direct	Only available in 1152 pins package
HSSL02	Direct	Only available in 1152 pins package
HSSL03	Direct	-

Check Packages chapter for more detail about each package.

HSSL IO pins

HSSL IO bank pads list [tab 15 - chap 7] details ULTRA300 HSSL bank pins.

Table 35: HSSL IO bank pads list [tab 15 - chap 7]			
PADS	Direction	Voltage (V)	Description
HSSL*_CKREFP	IN	HSSL*_TX_VDDA - 1.0 / HSSL*_VD- D1V8A – 1.8 / HSSL *__VDDCORE – 1.0	Differential positive REFCLK
HSSL*_CKREFN	IN	HSSL*_TX_VDDA - 1.0 / HSSL*_VD- D1V8A – 1.8 / HSSL*_VDDCORE – 1.0	Differential negative REFCLK
HSSL*_CALIBRATE	IN	HSSL*_TX_VDDA - 1.0 / HSSL*_VD- D1V8A – 1.8 / HSSL*_VDDCORE – 1.0	External reference resistor connection to calibrate TX/RX termination value
HSSL*_TX0P	OUT	HSSL*_TX_VDDA - 1.0 / HSSL*_VD- D1V8A – 1.8 / HSSL*_VDDCORE – 1.0	Transmit Data. SerDes dif- ferential positive output
HSSL*_TX1P	OUT	HSSL*_TX_VDDA - 1.0 / HSSL*_VD- D1V8A – 1.8 / HSSL*_VDDCORE – 1.0	Transmit Data. SerDes dif- ferential positive output
HSSL*_TX2P	OUT	HSSL*_TX_VDDA - 1.0 / HSSL*_VD- D1V8A – 1.8 / HSSL*_VDDCORE – 1.0	Transmit Data. SerDes dif- ferential positive output
HSSL*_TX3P	OUT	HSSL*_TX_VDDA - 1.0 / HSSL*_VD- D1V8A – 1.8 / HSSL*_VDDCORE – 1.0	Transmit Data. SerDes dif- ferential positive output
HSSL*_TX0N	OUT	HSSL*_TX_VDDA - 1.0 / HSSL*_VD- D1V8A – 1.8 / HSSL*_VDDCORE – 1.0	Transmit Data. SerDes dif- ferential negative output
HSSL*_TX1N	OUT	HSSL*_TX_VDDA - 1.0 / HSSL*_VD- D1V8A – 1.8 / HSSL*_VDDCORE – 1.0	Transmit Data. SerDes dif- ferential negative output
HSSL*_TX2N	OUT	HSSL*_TX_VDDA - 1.0 / HSSL*_VD- D1V8A – 1.8 / HSSL*_VDDCORE – 1.0	Transmit Data. SerDes dif- ferential negative output
HSSL*_TX3N	OUT	HSSL*_TX_VDDA - 1.0 / HSSL*_VD- D1V8A – 1.8 / HSSL*_VDDCORE – 1.0	Transmit Data. SerDes dif- ferential negative output
HSSL*_RX0P	IN	HSSL*_TX_VDDA - 1.0 / HSSL*_VD- D1V8A – 1.8 / HSSL*_VDDCORE – 1.0	Receive Data. SerDes differential positive input
HSSL*_RX1P	IN	HSSL*_TX_VDDA - 1.0 / HSSL*_VD- D1V8A – 1.8 / HSSL*_VDDCORE – 1.0	Receive Data. SerDes differential positive input
HSSL*_RX2P	IN	HSSL*_TX_VDDA - 1.0 / HSSL*_VD- D1V8A – 1.8 / HSSL*_VDDCORE – 1.0	Receive Data. SerDes differential positive input
HSSL*_RX3P	IN	HSSL*_TX_VDDA - 1.0 / HSSL*_VD- D1V8A – 1.8 / HSSL*_VDDCORE – 1.0	Receive Data. SerDes differential positive input

PADS	Direction	Voltage (V)	Description
HSSL*_RX0N	IN	HSSL*_TX_VDDA - 1.0 / HSSL*_VD- D1V8A - 1.8 / HSSL*_VDDCORE - 1.0	Receive Data. SerDes differential negative input
HSSL*_RX1N	IN	HSSL*_TX_VDDA - 1.0 / HSSL*_VD- D1V8A - 1.8 / HSSL*_VDDCORE - 1.0	Receive Data. SerDes differential negative input
HSSL*_RX2N	IN	HSSL*_TX_VDDA - 1.0 / HSSL*_VD- D1V8A - 1.8 / HSSL*_VDDCORE - 1.0	Receive Data. SerDes differential negative input
HSSL*_RX3N	IN	HSSL*_TX_VDDA - 1.0 / HSSL*_VD- D1V8A - 1.8 / HSSL*_VDDCORE - 1.0	Receive Data. SerDes differential negative input

Unused HSSL IO bank

When an HSSL bank is not used:

- HSSL dedicated power supplies can be left floating or grounded
- Other HSSL pins can be left floating (non-connected) or grounded

7.2.5 ADC and DAC IO banks

ULTRA300 has ADC and DAC hardware features. On the bigger 1152 pins package, ADC and DAC interfaces have dedicated ADC/DAC IO banks. On the smaller 484 pins package, ADC and DAC interfaces are shared with Direct banks.

ADC/DAC IO banks operate with dedicated 1.8V power supplies. These power supplies are independent from user bank voltages. In case of shared IO pins in the 484 packages, shared Direct bank (IOB12, IOB13 and IOB14) can operate in a different voltage as ULTRA300 IO architecture protects ADC/DAC shared IO to propagate IO glitches from Direct bank voltages to ADC/DAC power domains during power sequences. To ensure this behavior, recommended power sequencing must be applied.

ADC/DAC pin locations are package dependent and thus are detailed in the Packages chapter.

8. Packages

ULTRA300 FPGA is available in 27x27mm FBGA484, 45x45mm FBGA1152 and Daisy Chain packages.

This chapter details the characteristics of each package.

8.1 Package FBGA484

8.1.1 Package Overview

FBGA484 package features a subset of available die peripherals as presented in [FBGA484 features \[tab 1 - chap 8\]](#).

Table 36: FBGA484 features [tab 1 - chap 8]

IO Type	IO Bank organization (BankNb x PinNb)	IO Total Count
Complex IOs	4x34	136
Direct IOs	3x24+1x16+1x15 ⁽¹⁾⁽²⁾	103
Service IOs	48 ⁽³⁾	48
ADC	4x4 ⁽²⁾	16
DAC	4x8 ⁽²⁾	32
HSSL ⁽⁴⁾	1 x Quad	4 HSSL RX 4 HSSL TX 1 diff ref clock 1 calibration pin

- (1)** IOB08 has only first 15 pads connected in FBGA484 pins package
- (2)** Top direct bank IO are shared between direct pads and ADC/DAC pads in FBGA484 pins package:
- (3)** Some Service Bank IOs are not connected in FBGA484 pins package
- (4)** HSSL is currently under characterization – HSSL feature will be described in a later update of the Datasheet

8.1.2 Package Pinout and CAD libraries

Full package pinout is available in a dedicated Pinout Table document.

To help PCB designers, some CAD libraries are available in a dedicated archive.

Please refer to the ULTRA300 Documents chapter for the corresponding references.

8.1.3 FBGA484 Package Limits and Shared Pins

As described in this document, FBGA484 package does not have all IO feature available.

Users must be careful about IO when using this package. FBGA484 package limits are:

- CLK_OSC configuration pin is **NOT** available, this clock source cannot be used
- POK (Power OK) status pin is **NOT** available, power monitoring must be done by user
- SpaceWire configuration interface is **NOT** available
- Less Complex and Direct IO banks are available compared to FBGA1152 package
- IOB08 has only 15 pins available (instead of 16 in FBGA1152 package)
- ADC and DAC features are sharing IO with top Direct banks 12, 13 and 14 (less Direct IO are available if ADC/DAC features are used).
 - [FBGA 484 shared IO between direct banks and ADC/DAC \[tab2-chap 8\]](#) details ADC/DAC and Direct Bank IO sharing

Table 37: FBGA 484 shared IO between direct banks and ADC/DAC [tab 2 - chap 8]

ADC	DAC	DirectBank	IOSharing	Package NetName	BGAPIN
ADC_VDD1V8A		-	No	ADC_VDD1V8A	L5
ADC_VDD1V8A		-	No	ADC_VDD1V8A	N5
ADC00_IN0		IOB12	Yes	IOB12_D14	V4
ADC00_IN1		IOB12	Yes	IOB12_D17	T4
ADC00_IN2		IOB12	Yes	IOB12_D20	R3
ADC00_IN3		IOB12	Yes	IOB12_D15	U3
ADC00_IN4		IOB12	Yes	IOB12_D18	T3
ADC00_IN5		IOB12	Yes	IOB12_D21	P4
ADC00_IN6		IOB12	Yes	IOB12_D22	P3
ADC00_IN7		IOB12	Yes	IOB12_D23	P2
ADC01_IN0		IOB12	Yes	IOB12_D05	Y1
ADC01_IN1		IOB12	Yes	IOB12_D07	V1
ADC01_IN2		IOB12	Yes	IOB12_D12	V2
ADC01_IN3		IOB12	Yes	IOB12_D02	W2
ADC01_IN4		IOB12	Yes	IOB12_D09	T1
ADC01_IN5		IOB12	Yes	IOB12_D16	U4
ADC01_IN6		IOB12	Yes	IOB12_D10	T2
ADC01_IN7		IOB12	Yes	IOB12_D19	R2
ADC02_IN0		IOB14	Yes	IOB14_D12	B4
ADC02_IN1		IOB14	Yes	IOB14_D08	D3
ADC02_IN2		IOB14	Yes	IOB14_D07	C4
ADC02_IN3		IOB14	Yes	IOB14_D14	F1
ADC02_IN4		IOB14	Yes	IOB14_D10	C3
ADC02_IN5		IOB14	Yes	IOB14_D09	E3
ADC02_IN6		IOB14	Yes	IOB14_D20	A3
ADC02_IN7		IOB14	Yes	IOB14_D11	E2

ADC03_IN0	IOB14	Yes	IOB14_D19	B2
ADC03_IN1	IOB14	Yes	IOB14_D13	D2
ADC03_IN2	IOB14	Yes	IOB14_D15	C2
ADC03_IN3	IOB14	Yes	IOB14_D18	F2
ADC03_IN4	IOB14	Yes	IOB14_D17	D1
ADC03_IN5	IOB14	Yes	IOB14_D16	B3
ADC03_IN6	IOB14	Yes	IOB14_D21	E1
ADC03_IN7	IOB14	Yes	IOB14_D22	A2
DAC_VDD1V8A	-	No	DAC_VDD1V8A	G5
DAC_VDD1V8A	-	No	DAC_VDD1V8A	J5
DAC00_COMPIN0	IOB13	Yes	IOB13_D03	M3
DAC00_COMPIN1	IOB13	Yes	IOB13_D02	N1
DAC00_COMPIN2	IOB13	Yes	IOB13_D00	L1
DAC00_COMPIN3	IOB13	Yes	IOB13_D01	M1
DAC00_OUT	-	No	DAC00_OUT	H4
DAC01_COMPIN0	IOB13	Yes	IOB13_D05	J2
DAC01_COMPIN1	IOB13	Yes	IOB13_D04	K1
DAC01_COMPIN2	IOB13	Yes	IOB13_D09	N3
DAC01_COMPIN3	IOB13	Yes	IOB13_D06	K2
DAC01_OUT	-	No	DAC01_OUT	J4
DAC02_COMPIN0	IOB13	Yes	IOB13_D10	J1
DAC02_COMPIN1	IOB13	Yes	IOB13_D11	L3
DAC02_COMPIN2	IOB13	Yes	IOB13_D08	M2
DAC02_COMPIN3	IOB13	Yes	IOB13_D07	L2
DAC02_OUT	-	No	DAC02_OUT	G4
DAC03_COMPIN0	IOB13	Yes	IOB13_D14	M4
DAC03_COMPIN1	IOB13	Yes	IOB13_D15	N4
DAC03_COMPIN2	IOB13	Yes	IOB13_D13	L4
DAC03_COMPIN3	IOB13	Yes	IOB13_D12	K3
DAC03_OUT	-	No	DAC03_OUT	K4

8.1.4 Package Ball Map

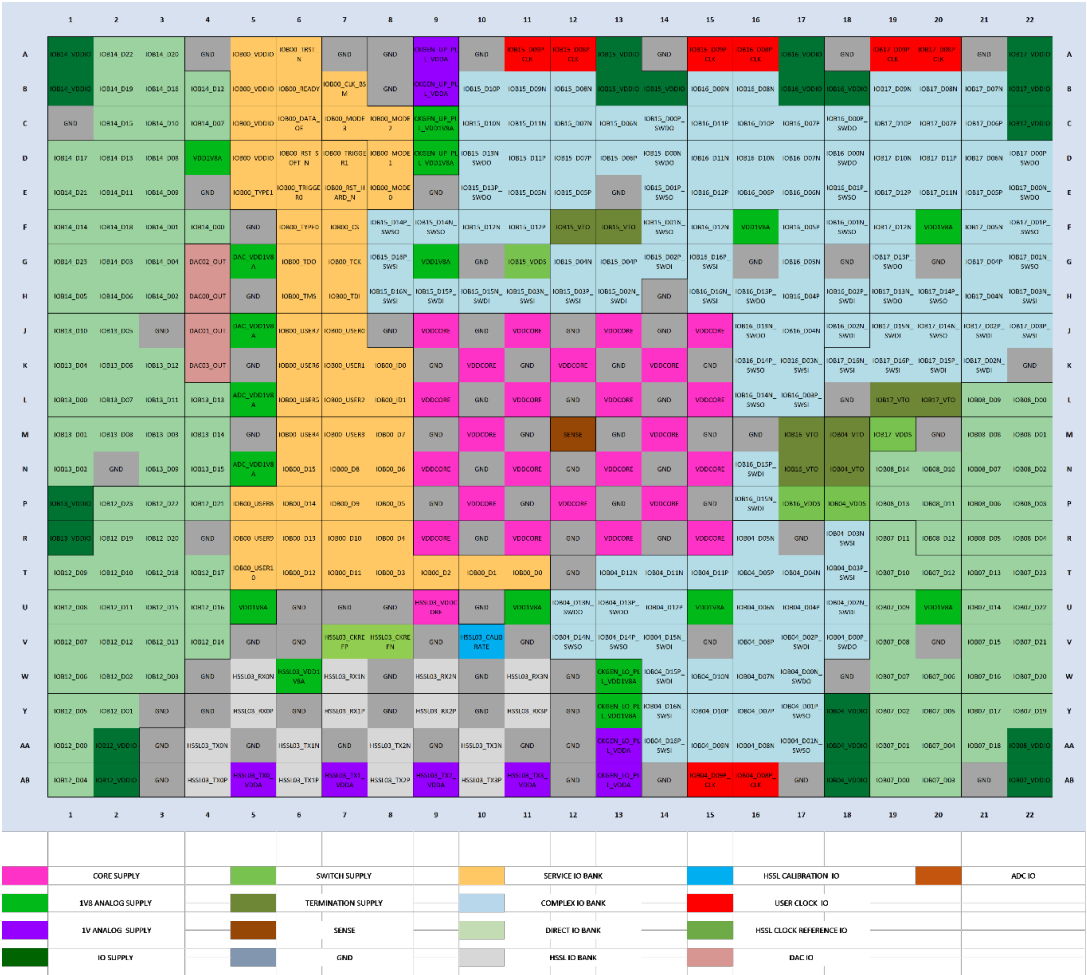


Figure 21: FF484 Ball Map [fig 1 – chap 8]

8.1.5 Mechanical Dimensions

TITLE: FCTEBGA 27X27 484 P1 B0.64 (1)
PACKAGE CODE: C05C (C Zero 5C)
JEDEC REFERENCE NUMBER: DESIGN REGISTRATION 4.14

DIMENSION (1) DATABOOK				
SYMBOL	MIN.	NOM.	MAX.	NOTE
A	2.86	3.06	3.25	2, 3
A1	0.43	0.53	0.63	4
A2	-	2.523	-	
b	0.55	0.65	0.75	6
D	27.00 BSC			7
D1	21.00 BSC			7
D2	-	26.60	-	
D3	-	2.50	-	
E	27.00 BSC			
E1	21.00 BSC			
E2	-	26.60	-	
E3	-	2.50	-	
e	1.00 BSC			7, 10
N	484			11
SD	0.50 BSC			7, 12
SE	0.50 BSC			

SYMBOL	TOLERANCE OF FORM AND POSITION (1) DATABOOK	NOTE
aaa	0.20	13
ccc	0.35	
ddd	0.20	
eee	0.25	
fff	0.10	

Figure 22: FF484 Package Dimensions 1 [fig 2 - chap 8]

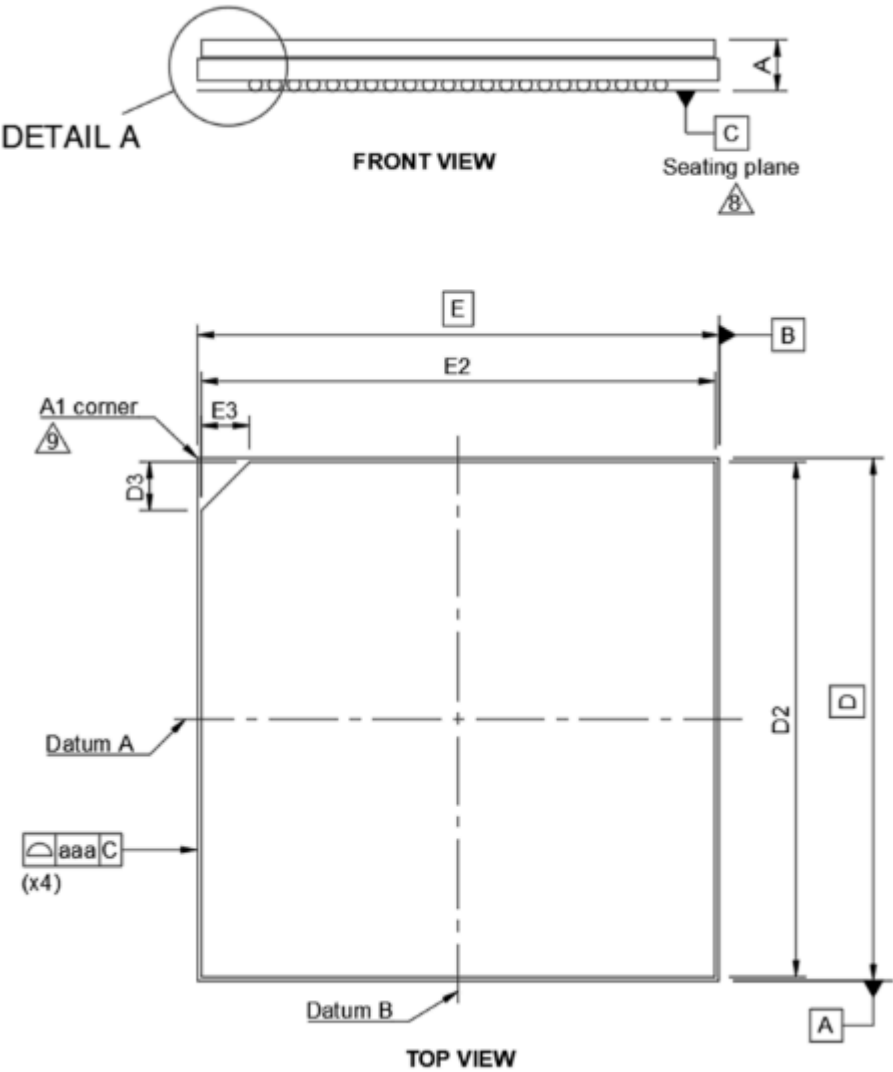


Figure 23: FF484 Package Dimensions 2 [fig 3 - chap 8]

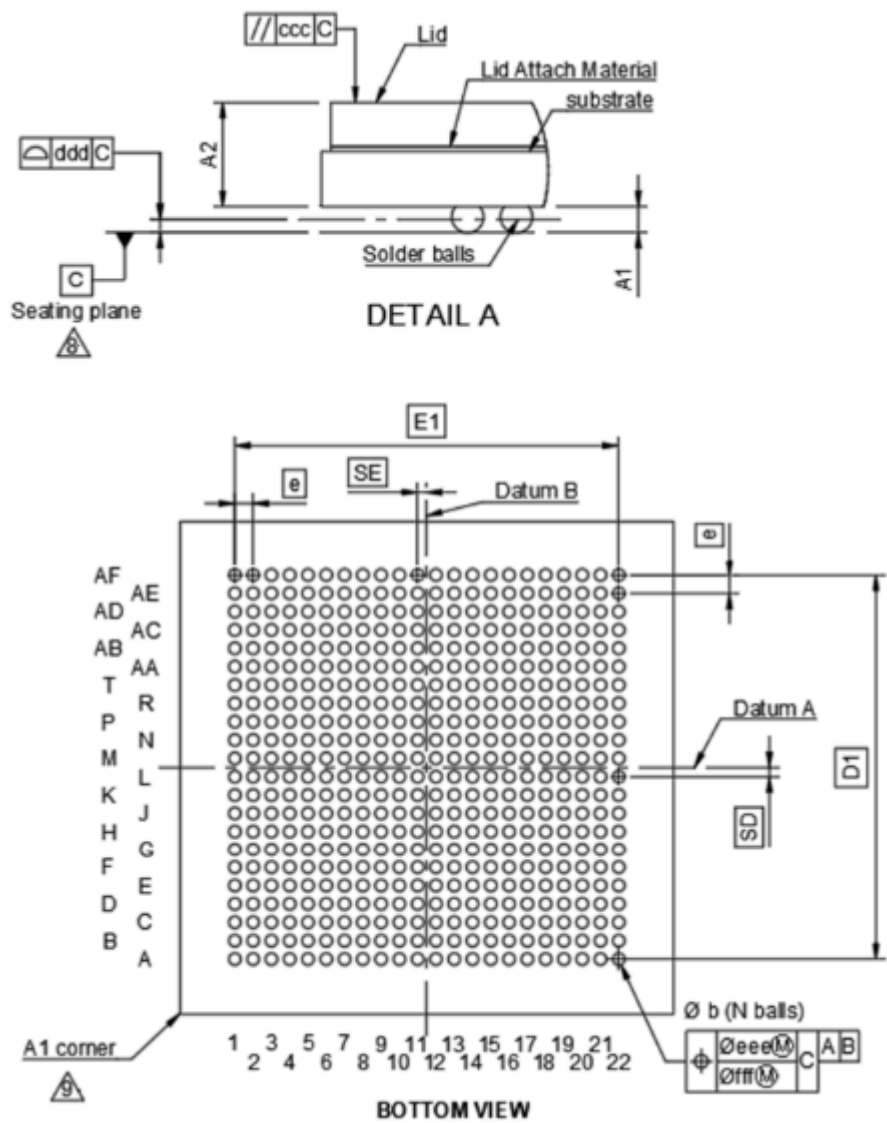


Figure 24: FF484 Package Dimensions 3 [fig 4 - chap 8]

NOTES

- 1. All dimensions are in millimetres. Dimensioning and tolerancing schemes are conform to ASME Y14.5M-2018 apart European projection.
- 2. FCTEBGA stands for Flip Chip Thermal Enhanced Ball Grid Array package.
- 3. The profile height, A, is the distance from the seating plane to the highest point on the package. It is measured perpendicular to the seating plane.
- 4. A1 is defined as the distance from the seating plane to the lowest point on the package body for BGA or the distance from the seating plane to the land for LGA.
- 5. Raw ball diameter before soldering on the package.
- 6. Dimension b is measured at the maximum diameter of the terminal (ball, land) in a plane parallel to Datum C.
- 7. BSC stands for BASIC dimensions. It corresponds to the nominal value and has no linear tolerance. For tolerances refer to form and position table. On the drawing these dimensions are framed.
- 8. Datum C (seating plane) is defined by the plane established by contact of the package bottom in three or more non-collinear areas (balls...) that support the freestanding package when it is placed on top of a planar surface.
- 9. The A1 corner is identified on the top surface of the package by using a marking or a physical feature. A distinguish feature is allowable on the bottom surface of the package to identify the A1 corner. Exact shape of each corner is optional.
- 10. e represents the solder balls/lands grid pitch(es).
- 11. N represents the total number of balls/lands.
- 12. Basic dimensions SD & SE are defining the balls/lands matrix position with respect to datums A and B.
- 13. For the tolerance of form and position definitions see below table.

SYMBOL	DEFINITION
aaa	The profile tolerance controlling the location and orientation of the sides of the package.
ccc	The parallelism tolerance controlling the orientation of the top surface of the package with respect to Datum C.
ddd	The profile tolerance controlling the location of the crowns of the balls/lands with respect to Datum C.
eee	The tolerance of position that controls the location of the pattern of balls/lands with respect to Datum's A and B.
fff	The tolerance of position that controls the location of the balls/lands within the matrix with respect to each other.

Figure 25: FF484 Package Dimensions 4 [fig 5 - chap 8]

8.2 Package FBGA1152

8.2.1 Package Overview

FBGA1152 package includes full die peripherals as presented in [FBGA1152 features \[tab 3 - chap 8\]](#).

Table 38: FBGA1152 features [tab 3 - chap 8]

IO Type	IO Bank organization (BankNb x PinNb)	IO Total Count
Complex IOs	10x34	340
Direct IOs	6x34+1x24	160
Service IOs	57	57
ADC	4x4	16
DAC	4x8	32
HSSL ⁽¹⁾	4 Quads	16 HSSL RX 16 HSSL TX 4 diff ref clocks 4 calibration pins

(1) HSSL is currently under characterization – HSSL feature will be described in a later update of the Datasheet

8.2.2 Package Pinout and CAD libraries

Full package pinout is available in a dedicated Pinout Table document.

To help PCB designers, some CAD libraries are available in a dedicated archive.

Please refer to the ULTRA300 Documents chapter for the corresponding references.

8.2.3 Ball Map

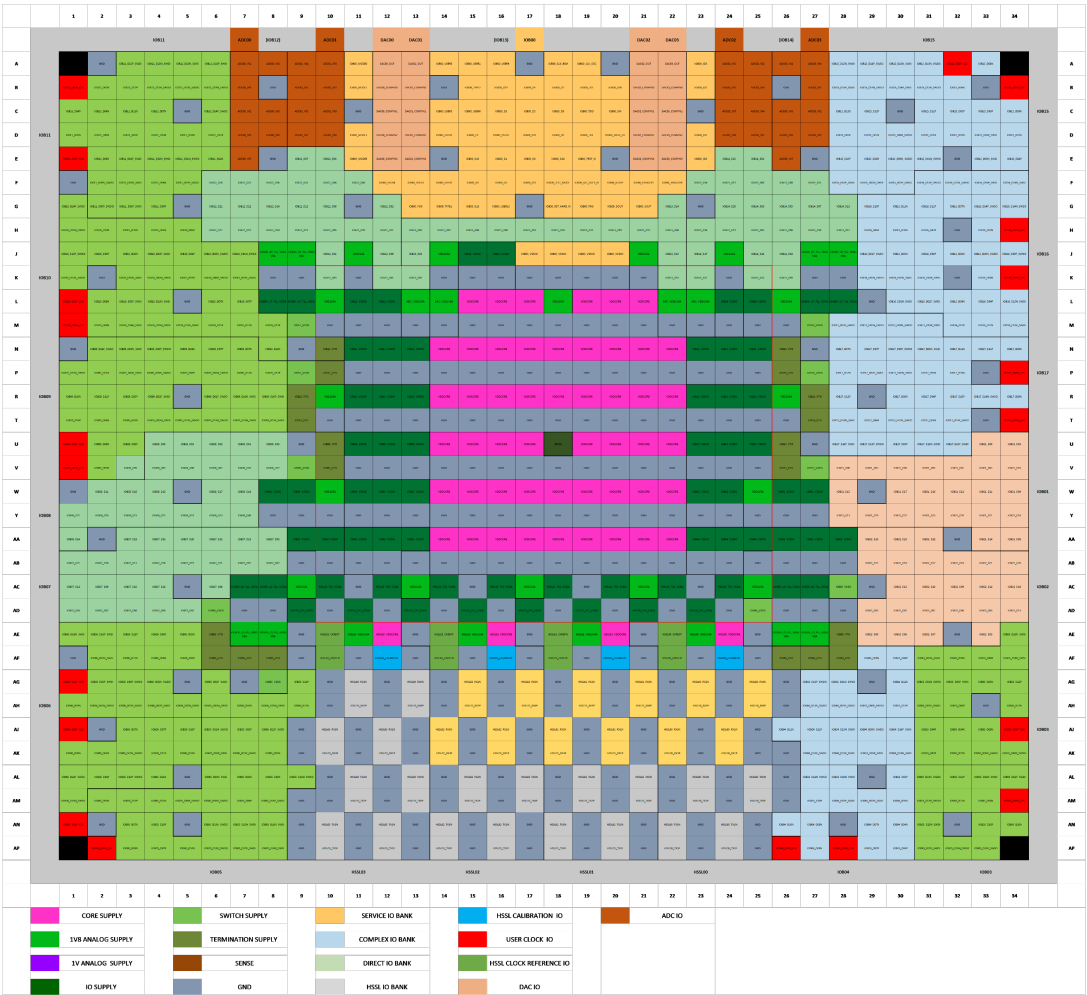


Figure 26: FF1152 Ball Map [fig 6 - chap 8]

8.2.4 Mechanical Dimensions

JEDEC REFERENCE NUMBER: DESIGN REGISTRATION 4.14

DIMENSION (1) DATABOOK				
SYMBOL	MIN.	NOM.	MAX.	NOTE
A	2.69	2.69	3.07	2, 3
A1	0.43	0.53	0.53	4
A2	-	2.35	-	
b	0.55	0.65	0.75	6
D	35.00 BSC			7
D1	33.00 BSC			7
D2	-	34.60	-	
D3	-	2.50	-	
E	35.00 B S C			
E1	33.00 BSC			
E2	-	34.60	-	
E3	-	2.50	-	
e	1.00 BSC			7, 10
N	1152			11
SD	0.50 BSC			7, 12
SE	0.50 BSC			

SYMBOL	TOLERANCE OF FORM AND POSITION (1) DATABOOK	NOTE
aaa	0.20	13
ccc	0.35	
ddd	0.20	
eee	0.25	
fff	0.10	

Figure 27: FF1152 Package Dimensions 1 [fig - chap 8]

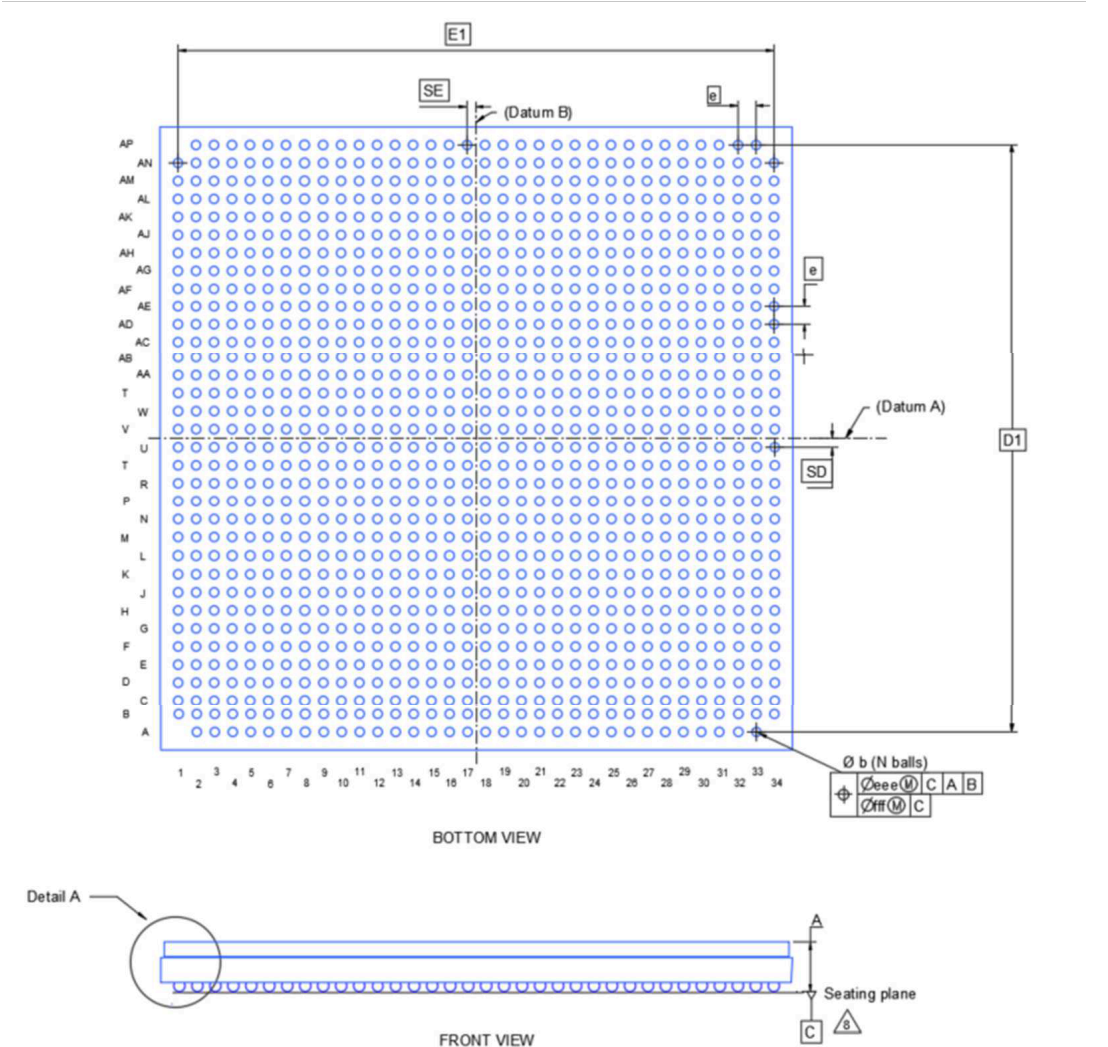


Figure 28: FF1152 Package Dimensions 2 [fig 8 - chap 8]

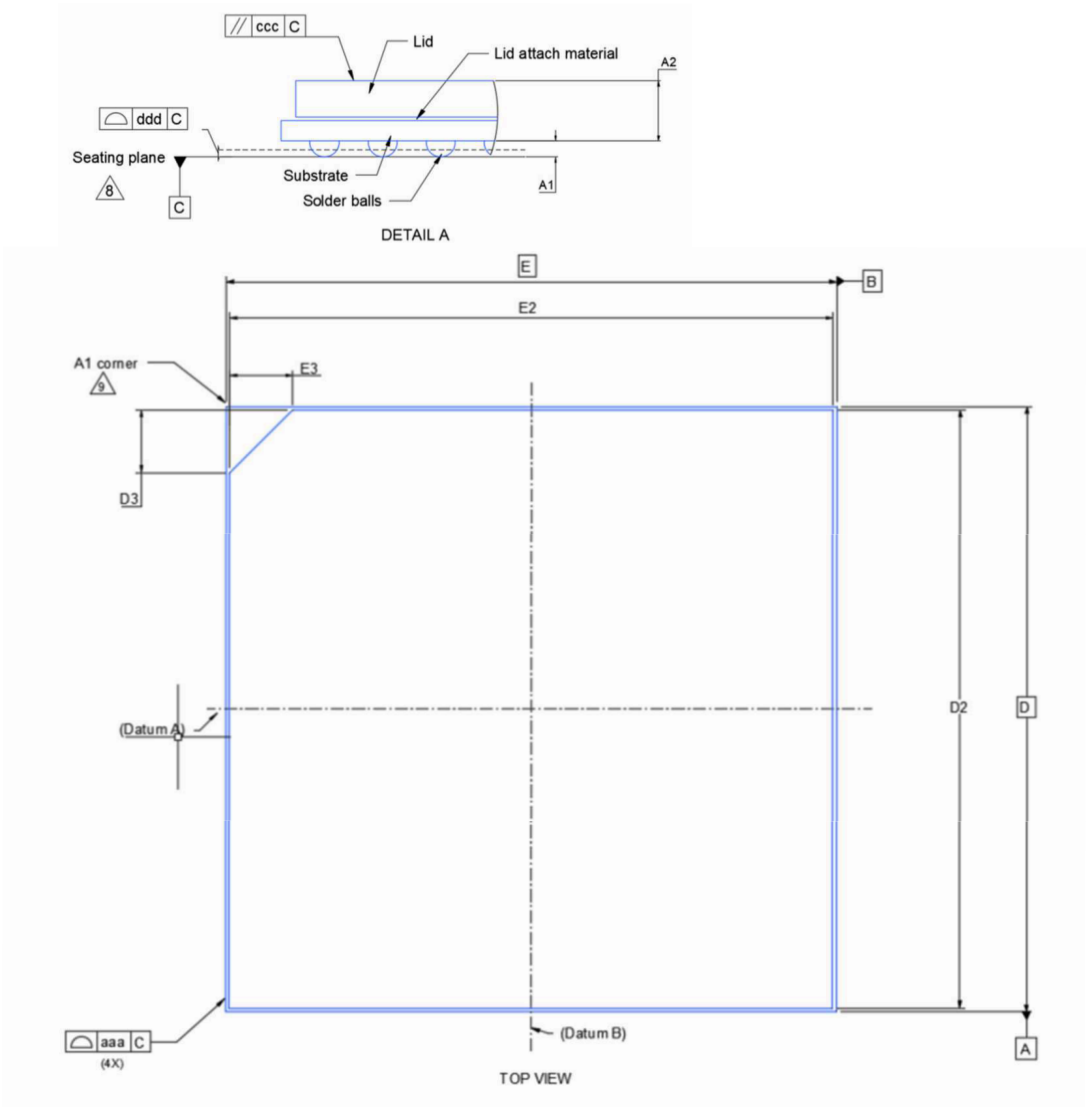


Figure 29: FF1152 Package Dimensions 3 [fig - chap 8]

NOTES

- 1. All dimensions are in millimetres. Dimensioning and tolerancing schemes are conform to ASME Y14.5M-2018 apart European projection.
- 2. FCTEBGA stands for Flip Chip Thermal Enhanced Ball Grid Array package.
- 3. The profile height, A, is the distance from the seating plane to the highest point on the package. It is measured perpendicular to the seating plane.
- 4. A1 is defined as the distance from the seating plane to the lowest point on the package body for BGA or the distance from the seating plane to the land for LGA.
- 5. Raw ball diameter before soldering on the package.
- 6. Dimension b is measured at the maximum diameter of the terminal (ball, land) in a plane parallel to Datum C.
- 7. BSC stands for BASIC dimensions. It corresponds to the nominal value and has no linear tolerance. For tolerances refer to form and position table. On the drawing these dimensions are framed.
- 8. Datum C (seating plane) is defined by the plane established by contact of the package bottom in three or more non-collinear areas (balls...) that support the freestanding package when it is placed on top of a planar surface.
- 9. The A1 corner is identified on the top surface of the package by using a marking or a physical feature. A distinguish feature is allowable on the bottom surface of the package to identify the A1 corner. Exact shape of each corner is optional.
- 10. e represents the solder balls/lands grid pitch(es).
- 11. N represents the total number of balls/lands.
- 12. Basic dimensions SD & SE are defining the balls/lands matrix position with respect to datums A and B.
- 13. For the tolerance of form and position definitions see below table.

SYMBOL	DEFINITION
aaa	The profile tolerance controlling the location and orientation of the sides of the package.
ccc	The parallelism tolerance controlling the orientation of the top surface of the package with respect to Datum C.
ddd	The profile tolerance controlling the location of the crowns of the balls/lands with respect to Datum C.
eee	The tolerance of position that controls the location of the pattern of balls/lands with respect to Datum's A and B.
fff	The tolerance of position that controls the location of the balls/lands within the matrix with respect to each other.

Figure 30: FF1152 Package Dimensions 4 [fig 10 - chap 8]

8.3 Daisy Chain Packages

Ultra300 RH FPGA is available in daisy chain packages.

The purpose of daisy chain packages is to offer solutions for testing and validate board assembly and soldering process without relying on functional prototypes or flight-quality components.

Daisy chain packages are designed to match commercial and space ULTRA300 version packages and share identical specifications:

- Materials (metal lid, glue, TIM, substrate, solder bumps and balls, etc...)
- Structure and density (substrate and die)
- FBGA footprint

Daisy chain patterns are implemented on substrate and die parts allowing first (bumps) and second (BGA balls) assembly levels testing and validation. These patterns are fixed and cannot be changed or reprogrammed by user.

ULTRA300 Daisy chain package exists in both 484 and 1152 pins versions.

ULTRA300

Daisy chain packages characteristics are detailed in a dedicated document.

Please refer to the ULTRA300 Documents chapter for the corresponding reference.

8.4 Space and Commercial ULTRA300 Package Compatibility

Space ULTRA300 RH and Commercial ULTRA300 packages share the same footprint. Thus, both version packages are pin to pin compatible.

9. Ordering code information

ULTRA300 RH is available in different grades:

- Prototype
- Engineering model
- Equivalent Flight model
- Flight Model ESCC 9030 Class 1
- Daisy Chain

ULTRA300 RH part number decoding [fig 1 - chap 9] details ULTRA300 RH part number.

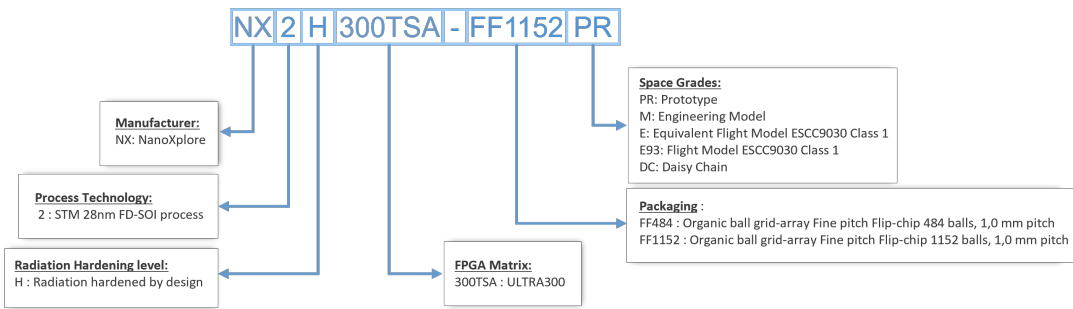


Figure 31: ULTRA300 RH part number decoding [fig - chap 9]

ULTRA300 RH Device List [tab 1 - chap] lists available ULTRA300 RH devices.

Table 39: ULTRA300 RH Device List [tab 1 - chap 9]

Part Number	Package	Temperature range	Quality level
NX2H300TSA-FF484PR	FBGA-484	+25°C	Prototype
NX2H300TSA-FF484M	FBGA-484	-40°C - +125°C	Engineering Model
NX2H300TSA-FF484E	FBGA-484	-40°C - +125°C	Equivalent Flight Model
NX2H300TSA-FF484E93	FBGA-484	-40°C - +125°C	Flight Model
NX2H300TSA-FF484DC	FBGA-484	+25°C	Daisy Chain
NX2H300TSA-FF1152PR	FBGA-1152	+25°C	Prototype
NX2H300TSA-FF1152M	FBGA-1152	-40°C - +125°C	Engineering Model
NX2H300TSA-FF1152E	FBGA-1152	-40°C - +125°C	Equivalent Flight Model
NX2H300TSA-FF1152E93	FBGA-1152	-40°C - +125°C	Flight Model
NX2H300TSA-FF1152DC	FBGA-1152	+25°C	Daisy Chain

10. Glossary

Acronym	Description
ADC	Analog to Digital Converter
AES	Advanced Encryption Standard
ALU	Arithmetic Logic Unit
CG	Clock Generator
CGA	Column Grid Array
CKG	Clock Generator
CMIC	Configuration Memory Integrity Check
CMOS	Complementary Metal Oxide Semiconductor
DAC	Digital to Analog Converter
DDR	Double Data Rate
DFF	D-Flip Flop
DPRAM	Dual-Port Read Access Memory
DPREG	Dual-Port Register
DSP	Digital Signal Processor
ECC	Error Correction Circuit
ECSS	European Cooperation for Space Standardization
EDAC	Error Detection And Correction
FPGA	Field Programmable Gate Array
FPU	Floating-Point Unit
HSSL	High Speed Serial Link
HSTL	High Speed Transceiver Logic
IOB	Input Output Block
LGA	Land Grid Array
LUT	Look-Up Table
LVCMOS	Low Voltage CMOS
LVDS	Low Voltage Differential Signal

CQFP	Ceramic Quad Flat Package
MOSI	Master Output Slave Input
MISO	Master Input Slave Output
OTP	One Time Programmable
PCB	Printed Circuit Board
PCI	Peripheral Component Interconnect
PCS	Physical Coding Sublayer
PHY	Physical (layer)
PLL	Phase-Locked Loop
PMA	Physical Medium Attachment
RH	Radiation Hardened
RHBD	Radiation Hardened By Design
SDR	Single Data Rate
SECDED	Single Error Correction, Double Error Detection
SEFI	Single-Event Functional Interrupt
SEL	Single-Event Latch-up
SER	Soft-Error Rate
SET	Single-Event Transient
SEU	Single-Event Upset
SIP	System in Package
SPI	Serial Peripheral Interface
SPREG	Simple-Port Register
SpW Tx & Rx	SpaceWire Transceiver & Receiver
SSTL	Stub Series Terminated Logic
SWaP	Size, Weight and Power
TMR	Triple Modular Redundancy
WFG	Wave Form Generator

11. Revision History

Version	Date	Revision
V1.0.0	July-2026	First Release